

Outsourcing Education:
Experimental Evidence from Liberia.
A Replication Study of Romero, Sandefur, and Sandholtz
(American Economic Review, 2020)*

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Abstract

Romero et al. (2020) examines the Liberian government’s partnership with private providers to manage a set of public schools. Privately run schools performed 0.18σ higher in student test scores for English and mathematics after one academic year. This public-private partnership was a cost-effective means of improving student learning compared to a counterfactual of public administration for all schools. We attempt to computationally reproduce these results and test their sensitivity through a robustness reproduction. After a failed computational reproduction of the results, we successfully reproduce the two headline results through a revised reproduction. We test the robustness of results by re-analyzing the test score findings alternating analytical choices through a multiverse framework of alternative researcher decisions. After altering treatment definition, control variable sets, imputation of missing observations, model selection, and outcome definitions, we estimate 320 re-analyses. We find that the results are robustly statistically significant. This reproduction endorses the claim that public-private partnerships can improve student learning with the caveat that improvements may be smaller than originally reported.

KEYWORDS: Reproducibility, Public-private Partnerships, Education, Outsourcing

JEL CODES: H41, I21, I28, O15, Z00

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1 Introduction

Reproductions and replications are considered to be an essential diagnostic tool for determining which bodies of evidence are considered reliable for use outside of academia and add to the foundations of existing literature within academia (King, 1995; Maniadis et al., 2017; Moonesinghe et al., 2007; Peterson and Panofsky, 2021). To improve our body of evidence on education interventions in developing settings, we attempt to reproduce Romero et al. (2020). This American Economic Review study implements a public-private partnership (the PSL program) for administration of schools among selected schools in Liberia by the Ministry of Education. Schools were first matched based on infrastructure and geography. Then, schools were assigned to treatment and control status using matched pair fixed effects. Additionally, they track a baseline sample of students prior to treatment assignment avoiding bias from selection by parents and teachers on the PSL program (potentially upwardly biasing the effect of the program). The authors of this study report that privately run schools performed 0.18σ higher in student test scores for English and mathematics after one academic year (the primary outcomes of interest). The student test scores are constructed using Item Response Theory (IRT)¹. This best practice in the education literature adds to the construct validity that their survey measures of test scores accurately capture student performance. They also provide evidence that this intervention was cost-effective to administer for the Liberian government compared to a counterfactual of public administration for all schools. From these findings, the authors highlight the importance of contract rules and procurement agreements in contracting school administration to the private sector in developing countries. The external validity of these results are limited by the bilateral agreements that the Liberian government used to procure contracts as opposed to a competitive bidding process. Given the expanding role of public-private partnerships adoption amongst developing countries (in particular in Sub-Saharan Africa), it is valuable to determine the reproducibility and robustness of these results.

Using the publicly provided replication packet for this study, we attempt to computationally reproduce the results presented in this study. Some of the results are unreproducible due to data restrictions. Notably, due to data restrictions the attractive findings for cost-effectiveness are unreproducible and therefore unverifiable across sensitivity checks. Beyond data restrictions, we are unable to computationally reproduce the presented results using the replication packet as provided. The primary error appears to derive from a Stata command call for ‘xtivreg’ within the ‘reghdfe’ function. Rather, using modifications to the provided scripts we are able to successfully produce a revised reproduction of the headline results that student test scores for English and mathematics were higher for schools in the PSL public-private partnership. These outcome measures were pre-specified as primary outcomes of interest in the authors’ pre-analysis plan (Romero et al., 2017).

Further, we test the sensitivity of the presented results by implementing a series of re-analyses for these two main outcomes. Through the lens of a multiverse framework (Simmons et al., 2021), we test 15 alternative choices that the researchers may have taken across five analytical dimensions: the definition of treatment, the set of control variables used, the handling of missing observations, the model for treatment effects used, and the definition of measuring the outcomes. Using the cross-combination of these choices, we estimate 320 re-analyses – 160 for both English and mathematics test scores. We find that the results are consistently statistically significant and the magnitude of the results from the preferred specification by the original authors are well within the range of possible point estimates. 93% of re-analyses estimate a statistically significant estimate with the same sign as that originally reported. This provides confidence in the external validity of these findings for policy-makers to utilize public-private partnerships as administrators of publicly

¹Appendix Table A.1 details the main outcomes presented by the authors in Table 3 of original manuscript.

funded schools to improve student learning in developing countries.

This reproduction contributes to the Mertonian norms (Merten, 1942) for the practical application of the open science movement in contemporary scientific research as a validation tool for stakeholders and a correction tool for researchers. This study also adds to the growing literature on the reporting of reproduction studies to verify internal validity and assessing the external validity of influential findings in economics (Brodeur et al., 2024; Brodeur et al., 2025; Kjelsrud et al., 2023; McWay et al., 2024; Cong et al., 2024; Braaksma and McWay, 2025; McWay, 2025)

2 Reproducibility

2.1 Computational Reproduction

Using the replication package², we attempt to computationally reproduce the table results presented in the manuscript of Romero et al. (2020). The reproduction was conducted in Stata 16 – compatible with the original coding software Stata 14 used in the replication packet. This reproduction was initiated through the Robustness and Replicability in Economics (R2E) project (Rose et al., 2024).

We follow the standardized protocol³ developed by the Robustness and Replicability in Economics (R2E) to assess errors in the replication packet (Ankel-Peters et al., 2024). We was unable to computationally reproduce the results using the aforementioned replication packet. We recorded 127 errors⁴ which broke the code in attempting this reproduction over 8 of the 82 scripts provided⁵. Although, the authors state explicitly⁶ within the README description document of the replication packet that several scripts, while providing the code to reproduce results in the manuscript, will not run because of data restrictions related to personalized identifiable information necessary for the results. This limitation is primarily the result of preventing identification of operators for the PSL program and restricted educational data from the Liberian Ministry of Education. Considering this, 21 scripts are unable to be run⁷ and restricts the computational reproduction of four tables and figures⁸. This is clearly demarcated in the master scripts of the replication packet by preventing these programs from running by conditioning them as ‘public == 0’. This limitation mainly prevents the reproduction of the heterogeneity analysis related to cost-effectiveness for the public-private partnership between the Liberian government and NGO school operators. The inability to reproduce the results by construction does not weaken the claims of the authors made regarding the advantage of public-private partnerships as a cost-effective tool for improving student learning. But it does induce caution over whether these are robust results as we cannot perform a reproduction and subsequently a reproduction of these results to test their sensitivity to researcher decisions.

Beyond the scripts that are unreproducible due to data restrictions, we was also unable to computationally

²The replication packet (Romero et al., 2018) can be found at Harvard’s Dataverse <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/50PIYU>.

³Further details on this protocol can be found in Appendix Section A.

⁴For further details on the errors encountered in the computational reproduction, please reference Appendix Section B.

⁵Not all scripts provided are necessary to computationally reproduce the manuscript results. Some of the scripts relate to supplementary analysis not presented, and nearly half of scripts relate to endline survey data analysis which is an extension of the original analysis.

⁶From the README documentation file of the replication packet: “I would love to tell you all you have to do is run both master files and you are done, but the R and the STATA code intertwine a bit. In addition, there is PII data that we cannot share (to safeguard human subjects), so some of these codes will not run. This includes teacher IDs, school IDs, and operator identifiers.” (Romero et al., 2018)

⁷These scripts are expressed in Appendix Table A.2.

⁸These are Figures 1, 2, and 3 and Table 10 in the original manuscript.

reproduce the remaining results for this study. Of the 127 documented coding errors⁹ that prevents the computational reproduction of the presented results, the vast majority of these are related to an issue with updating the Stata package ‘reghdfe’ as implied by line 39 of the script ‘00_MasterFile.do’¹⁰. Resolving this update¹¹, the main regressions were unable to run due to errors¹² with calling the ‘xtivreg’ command through the ‘reghdfe’ command in user-created commands generated by the original authors in the script ‘TvsC_tables_ML.do’. Therefore, with the replication packet as written and following an honest and diligent attempt to understand the complexity of the functions called and error messages raised, we are unable to reproduce any of the main results provided in the manuscript.

This study is neither computationally reproducible from analytic data (CRA) nor computationally reproducible from raw data (CRR) as described by the Guide for Advancing Computational Reproducibility in the Social Sciences (ACRe) established by the Berkeley Initiative for Transparency in the Social Sciences (BITSS) (Bogdanoski et al., 2020). As such, we assign the computational reproducibility of Romero et al. (2020) as a Level 3¹³ within the BITSS 10 point scale. Adjusting the level to account for confidential raw data, we upgrade this to an adjusted Level 3 (L3*)¹⁴ score within the BITSS 10 point scale.

It is also worth noting that while the replication packet was quite thorough – providing both raw data, cleaning code, and final analysis code and data – the complexity of the data structure is a bit cumbersome and the primary analysis functions generated by the authors do not provide comments or details that allow replicators to easily troubleshoot through issues as they arise¹⁵. Additionally, while the scripts for cleaning and analysis are laid out well within the README description file, the datasets are not well documented. Rather, as the replicator we must parse through several dozen scripts to back out how datasets and scripts connect with one another. The replication packet meets the standard for a push to run analysis with a master script for computational reproduction, but – while unintentional – the difficulty in understanding the component parts may limit the replicability of this study for examination by others beyond the original authors. Additionally, it is not readily apparent that the midline dataset and analysis are the main results as there is included an extension for another wave of survey data for long-run treatment effects labeled endline. This could have been more clearly defined within the README document and made more expressly clear where the main manuscript results are produced using which dataset.

⁹A comprehensive account of unexpected coding errors are presented in Appendix Table A.3.

¹⁰Error message "function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);"

¹¹The issue arises from updating ‘reghdfe’ from ‘00_MasterFile.do’ by running line 39 ‘ssc install reghdfe, replace’. The solution is to run the following commands in order: (1) ‘cap ado uninstall ftools’, (2) ‘net install ftools, from("https://raw.githubusercontent.com/sergiocorreia/ftools/master/src/")’, (3) ‘cap ado uninstall reghdfe’, (4) ‘net install reghdfe, from("https://raw.githubusercontent.com/sergiocorreia/reghdfe/master/src/")’

¹²Error message "parentheses unbalanced r(132);"

¹³“Level 3 (L3): Analytic data and code are partially available, but raw data and cleaning code are missing. Possible improvements include: completing analysis data and/or code, adding raw data, and adding analysis code.” (Bogdanoski et al., 2020)

¹⁴“Adjusted Level 3 (L3*): All analysis code is provided, but only partial instructions to access the analysis data are available. This means that the original authors have provided some, but not all, of the following information: (a) Contact information, including name of the organization(s) that provides access to at least one individual’s data and contact information. (b) Terms of use, including licenses and eligibility criteria for accessing the data, if any. (c) Information on data files (meta-data), including the name(s) and number of files, file size(s), relevant file version(s), and number of variables and observations in each file. Though not required, other relevant information may be included, including a dataset dictionary, summary statistics, and synthetic data (fake data with the same statistical properties as the original data). (d) Estimated costs for access, including monetary costs such as fees and licenses required to access the data, and non-monetary costs such as wait times and specific geographical locations from where researchers need to access it.” (Bogdanoski et al., 2020)

¹⁵McWay et al. (2021) provides best practices for script structures in Stata to limit this necessary troubleshooting.

2.2 Revised Reproduction

To remedy the failed computational reproduction, we make an attempt to create a revised reproduction of the main outcomes of interest – student test scores for English and mathematics – presented in Table 3 of Romero et al. (2020). Using the script ‘12a_MidlineResults.do’ from lines 53 through 239 as a template, we reproduce these results in Table 1. Using a revised coding script for the primary outcomes of interest as identified by the pre-analysis plan (Romero et al., 2017), we are able to reproduce the intention to treat (ITT) results. But we were unable to resolve the instrumental variable (IV) regression call through ‘reghdfe’ for the command ‘xtivreg’ which allows for fixed effects in an IV specification which produces the treatment on treated (TOT) results with the correct standard errors. Instead, we present the closest estimated specification using fixed effects without the IV. These deviations are minor and do not significantly change the magnitude, direction, nor statistical significance of the TOT results. Therefore, we are able to reproduce the headline results of the manuscript within a degree of confidence. A summary detailing the provisions in the replication packet as well as the result of this revised computational reproduction are presented in Table 2.

The focus of this reproduction study is limited to the two primary outcomes of interest expressed in the pre-analysis plan by the authors. While we are unable to computationally reproduce the other results presented in the manuscript which are not limited by data restrictions, we are confident that other replicators interested in reproducing the secondary or heterogeneous analyses will be able to do so by creating revised, simplified versions of the code provided in the replication packet. Given a successful revised reproduction for student test scores in English and mathematics, we re-analyze the sensitivity of these results to researcher decisions in the analysis with a robustness reproduction.

3 Robustness Reproduction

Noting the successful reproduction of the study’s main findings, we examine the robustness reproduction¹⁶ of the study through re-analyses modified from the provided study’s data. We perform a series of re-analyses pre-registered on the Open Science Platform¹⁷. Pre-registering the re-analyses prior to analysis limits ad-hoc discernment in the reproduction process and increases the validity of using standard statistical significance as a measure of robustness for the original presented results (Brodeur et al., 2024; Arpinon and Espinosa, 2023; Simmons et al., 2021).

There are two outcomes of interest that this reproduction will focus on which are highlighted as the headline results of the study in the abstract of the original paper as well as specified as primary outcomes of interest in the authors’ pre-analysis plan. These are the standardized student test scores for English and mathematics conducted using Item Response Theory (IRT). The author’s preferred specification is the intention-to-treat (ITT) of a randomized controlled trial clustered across schools with matched pair fixed effects matched on school level characteristics and clustered robust standard errors at the school level (i.e., the level of treatment)¹⁸. The preferred results show a statistically significant improvement of 0.18σ in both English and math test scores for students in schools operated by the PSL public-private partnership in comparison to test scores of students in government run schools. We expound upon this specification in Subsection 3.1. A third outcome of interest which we cannot re-assess, yet equally prominent in the study,

¹⁶As defined by Dreber and Johannesson (2024)

¹⁷Re-analyses were pre-registered in a pre-analysis plan on July 16th, 2024 and can be found at: <https://osf.io/hxsuz>.

¹⁸This specification, along with some heterogeneity analysis, is pre-registered through the AEA RCT Registry which can be found at: <https://www.socialsciencesregistry.org/trials/1501/history/17622>.

is the efficiency-equity trade off. Due to confidential data pertaining to grade enrollment for the students, I am unable to reproduce the outcome¹⁹ that constructs the ‘maximum capacity’ constraint driving the efficiency-equity trade off.

While this specification is reasonable, Steegen et al. (2016) and Simonsohn et al. (2020) argue that the sensitivity of reported findings are largely dependent upon a cascade of researcher decisions. In this framework, there is a multiverse of potential ‘reasonable’ specifications that could be implemented. To assess the degree to which researcher decision making influences the reported results, we estimate a series of re-analyses for these two outcomes of interest as motivated by multiversal robustness checks. In this setting, we motivate in Subsection 3.2 several analytical dimensions which the original authors have degrees of freedom in selecting a preferred specification and may have implemented alternative decisions. To succinctly present the multiverse of resulting values, we summarize this information using specification curves (Simonsohn et al., 2020; Klein, 2024) and a sensitivity dashboard (Bensch et al., 2025). The specification curve sorts the resulting re-analyses by the size of the estimate and allows for comparison of alternative specifications against the preferred specification. The sensitivity dashboard displays summary metrics of re-analyses deviation from the original estimates.

3.1 Model Specification

The preferred model specification presented in “Section II. Experimental Results” of the original manuscript follows the pre-specified specification laid out in the pre-analysis plan (Romero et al., 2017). Equation 1a is the ITT specification with matched-pair fixed effects grouped by baseline school characteristics, and Equation 1b is a modification that allows for individual (X) and school (Z) level controls. The coefficient of interest is β_1 and β_2 for the two equations given treatment assignment ($treat$) of the PSL program to schools, respectively. Y are student test scores standardized against the distribution of scores in the control group which vary by the individual (i) student performance, treatment assigned school (s), and group (g) matched pairing across schools. English and mathematics test scores are pre-specified as the main outcomes of interest in this study. ε are robust standard errors clustered at the school level.

$$Y_{isg} = \alpha_g + \beta_1 treat_s + \varepsilon_{ig} \quad (1a)$$

$$Y_{isg} = \alpha_g + \beta_2 treat_s + \gamma_2 X_i + \delta_2 Z_s + \varepsilon_{ig} \quad (1b)$$

The test scores for English and mathematics are constructed using Item Response Theory (IRT) and are normalized relative to the control group’s test score performance. Therefore, test scores are measured in standard deviations about a normal distribution. Further reanalysis involves variations in researcher decisions related to this preferred specification.

The identification relies on the use of matched pair fixed effects to identify the impact of school level treatment effects. The matched pair fixed effects increases the precision of the treatment effects by creating common support about the distribution of the influence school characteristics have over student learning across treatment assignment. The main threat to this identification strategy is that some parents or teachers may select on schools that are contracted through the PSL program because they may have a high incentive to improve student learning or more resources to compensate or support teachers. This potential issue

¹⁹This result is reported result is in Table 5 Column 1 of Romero et al. (2020)

is resolved by tracking a pre-baseline sample of students prior to treatment assignment as the sample of analysis.

3.2 Multiverse of Re-analysis

In determining the multiverse of analytical paths, we assess the following analytical dimensions from which the original authors may have deviated from their preferred specification. These dimensions relate to the definition of treatment assignment, the set of control variables included in the specification, the imputation of missing observations, alternative models, and redefining the outcome measures. Within each analytical dimension, we consider various alternative choices. We consider two ways of defining treatment, four various sets of control variables, two choices for handling missing observations, two potential models to estimate treatment effects, and five alternative definitions of the outcome measures. Including the original selected choices, this results in 15 choices that might be combined to result in a chosen specification. To limit ad-hoc selection of these alternative choices and to allow the use of statistical significance as a measure of robustness, we register a pre-analysis plan on the Open Science Platform the choices we estimate under these analytical dimensions²⁰. A full list of the pre-specified analytical choices are presented in Appendix Table A.4.

Further, within the framework a multiverse of potential analysis paths we estimate the combination of all the potential choices in the analytical decision process. So, given the combination of the 15 feasible choices, this results in 320 multiversal re-analyses (160 for each outcomes). The specification curves in Figures 1 and 2 summarize the results of these re-analyses relative to the original specification for English test scores and mathematics test scores, respectively. Further details on each reanalysis performed can be found in Tables A.5 and A.6 in Appendix Section C for English and mathematics test scores, respectively.

Both specification curves show that the preferred specification fall centrally near the median estimated effect size amongst all possible analysis paths. Notably, for both outcomes the combination of choices result in almost entirely statistical significant results. None of the analysis paths for analyzing English or mathematics test scores switches signs for the treatment effect. These specification curves show a robustness of the preferred specification against changes in analytical choices regarding the definition of treatment assignment, the set of control variables including in the specification, the imputation of missing observations, alternative models, and redefining the outcome measures. Now we elaborate on the influence of these five dimensions of analytical choice and the alternative choices the original authors may have reasonably chosen.

Analytical Dimension: Defining Treatment The original authors pre-specify Equation 1a in their pre-analysis plan to estimate the intention-to-treat (ITT) effect of the PSL program on student learning. But the authors also note that several of the schools that were assigned treatment assignment never received proper administration from providers of the PSL program. As such, there is non-compliance in treatment assignment due to treatment infidelity which warrants the estimation of treatment-on-the-treated (TOT) effects (a robustness check implemented by the authors). Under this analytical dimension, we vary the choice of treatment definition across (1) the ITT and (2) TOT definitions of treatment. Examining Figure 1 for English test scores and Figure 2 for math test scores, the TOT estimates are generally larger than that of the ITT treatment. This is sensible as the TOT estimates account for the downward bias of treatment infidelity by some providers.

²⁰The pre-analysis plan can be found here: <https://osf.io/hxsuz>

Analytical Dimension: Control Variables The original authors pre-specify in their pre-analysis plan two sets of baseline controls: a set of individual (student) level characteristics and a set of school-level attributes. None of the control variables appear to include collider bias. These controls are:

- Individual controls: Age, gender, wealth index, and school enrollment prior to baseline surveys.
- School controls: School enrollment rates prior to baseline surveys, baseline quality of facilities, an indicator for rural/urban status, and the distance to the nearest bank (a proxy for centrality and socioeconomic status of the school’s community).

There are three alternative reasonable variations that the authors might have selected as control measures. Considering the RCT design, the authors might have considered using no controls at all given randomization removes bias from selection on observables. Given the match-pair randomization design, the authors may have forgone school-level controls since they have already matched treatment assignment on nearest neighbors within the distribution of school attributes. And finally, the authors might have naively thought that only school-level attributes impact student learning. Under this analytical dimension, we vary the set of control sets across (1) the pre-specified set of controls, (2) no control variables, (3) only individual-level controls, and (4) only school-level controls. Examining Figure 1 for English test scores and Figure 2 for math test scores, both outcome measures are robust to all four sets of control variables. The treatment effects and statistical significance of the results are not strongly influenced by the use of control variables implying that treatment is largely uncorrelated with student or school characteristics.

Analytical Dimension: Handling Missing Observations The original authors pre-specify that they will handle missing observations of the control variables by imputing zero and including a dummy variable for missing status²¹. While sensible, the final non-response for control variables was quite low (6%). It is therefore reasonable that the authors might have forgone imputation of missing values in their estimation. Under this analytical dimension, we vary the choice of handling missing observations by (1) following the imputation method pre-specified by the authors and (2) making no imputation. Examining Figure 1 for English test scores and Figure 2 for math test scores, while there is a spread of the selected choice about the specification curve there are generally higher treatment effects when the preferred imputation method is used. This suggests that the bias from missing observations does tend to downwardly bias the results.

Analytical Dimension: Alternative Estimation Specification The original authors pre-specify a match-pair RCT design with clustered randomization of the PSL program by schools. Additionally, they pre-specify in their pre-analysis plan an ANCOVA design which controls for baseline levels of the outcome measures. This is sensible as it improves precision while maintaining statistical power. Though, the authors decide against this the presentation of this model specification due to a risk of attenuation bias from baseline imbalance in test scores²². Given both models are pre-specified, it is reasonable for the authors to have presented either in their preferred specification. Under this analytical dimension, we vary the choice of

²¹“To include all of our observations in the regressions and include student level controls, we will replace any missing values for the controls with zeroes. To account for this in our regression specification we will include an indicator for whether the baseline data is missing and interact this indicator with all the baseline controls.” (Romero et al., 2017)

²²“Adding controls, as in equation (2), should increase the precision of our results. However, controlling for baseline outcomes, as in equation (3), may also risk attenuation bias in the treatment effect estimates if the baseline outcomes are imbalanced. This is, in fact, what we observe in our baseline data, as shown in Table 8. For both English and math scores, students in treatment schools score higher than control schools (significant at the 5% level), by roughly 0.07 standard deviations in each case.” Romero et al. (2017)

model selection by (1) using the original RCT design, and (2) estimating an ANCOVA model. Examining Figure 1 for English test scores and Figure 2 for math test scores, the specification curves confirm the anticipated downward attenuation bias from the ANCOVA specification. This confirms the caution against the implementation of the ANCOVA model for this sample.

Analytical Dimension: Defining Outcomes The original authors rely on item response theory (IRT) to standardize test scores as their measures of student learning. They prefer this definition as it accounts for test taker ability, the difficulty of questions, and allows for comparison across grade levels and survey rounds. While this is a sensible definition, there are several alternative measures of student learning that could have been adopted from the education literature. First, some providers began treatment implementation concurrent with baseline surveys. So it is sensible to test the impact of student learning through IRT measures taken at baseline. Additionally, there may be large variation across grade level such that the IRT measures should be standardized within grades rather than across schools. Third, principal component analysis (PCA) is a useful means of summarizing factors predictive of test scores which may be more informative of student learning. And finally, the authors may have used the more traditional view of the percentage of correct answers on test scores. Under this analytical dimension, we vary the definition of the student learning by (1) using the original IRT measures, (2) using baseline levels of IRT, (3) standardizing IRT within grade levels, (4) estimating student through a PCA summary statistic, and (5) using the percentage of correctly answered questions on the tests. Examining Figure 1 for English test scores and Figure 2 for math test scores, there are two definitions of the outcome measure that influence the effect size and statistical significance of the results. For both English and math test scores, using the percentage of correctly answered questions on the tests leads to much smaller treatment effects. This may just be an artifact of the units used in this definition compared to the other four definitions (e.g., percentage points vs. standard deviations). Secondly, the English test scores when standardizing IRT by grade level are much larger treatment effects. This suggests that there are large differences in student understanding for English across grade level which is not present for mathematics.

3.3 Reproducibility Indicators

To summarize the replicability of these results as a consensus, we use the reproducibility indicators framework endorsed by Dreber and Johannesson (2023). This framework produces summary measures of the robustness of results across several metrics presented through the sensitivity dashboard in Figure 3 and summary statistics presented in Table 3. Appendix Section A.5 further details the definition and construction of the indicators in these two exhibits. The reproducibility indicators assessed are:

1. Significance Agreement Indicator: Measures the share of re-analyses that were both statistically significance in both the original results and re-analyses.
2. Relative Effect Size Indicator: Measures the mean/median of the re-analysis coefficients divided by the original coefficient.
3. Relative Significance Indicator: Measures the mean of the test statistics of the re-analyses divided by the original test statistic.
4. Effect Size Variation Indicator: Measures the standard deviation of the re-analysis coefficients divided by the original estimated standard error.

5. T-value Variation Indicator: Measures the standard deviation in the re-analysis test statistics.

Assessing these indicators, Table 3 informs us that 93.60% of all re-analyses remained statistically significant, and that the average re-analysis coefficient is smaller by a factor of 0.18 than the original coefficient estimate. This suggests that the reported effects are likely overstated but remain significant results. Further, the re-analysis test statistics are on average smaller by a factor of 0.96 the size of the original reported test statistic. And finally, the standard deviation of the re-analysis coefficients are a factor of 1.63 of the original reported standard error. This implies that the estimated effects are less precise than that original reported by the preferred specification.

Interpreting the sensitivity dashboard in Figure 3, only 7% of re-analyses were statistically insignificant at a 5% level with a relative decrease of 27% and 23% in the size of the p-value on average for English and math test scores, respectively. The median re-analysis coefficient is 85% smaller for English test scores and 78% smaller for math test scores than the original estimate reported. This is concurrent evidence that while the results are robustly statistically significant the reported effect size is likely inflated by the model specification.

3.4 Pre-analysis Plan

This study pre-registers the model specification, the sample of analysis, treatment identification, the set of control variables, and the outcomes of interest in a pre-analysis plan (PAP) registered through the AEA RCT Registry (Romero et al., 2017). This is a very comprehensive PAP expounding on empirical methodology, intervention implementation by partners, baseline characteristics, an intervention timeline, clearly stated hypotheses, and defined outcome measures mapped to the survey instrument categorized by families of outcomes. They even go so far as to consider heterogeneous treatment effects.

The adherence to this PAP is very well maintained in the published manuscript. The authors follow through with the use of the pre-specified model and controls. And construct outcome measures as described in the PAP. A small deviation is that they only report Equation 3 of the PAP (i.e., the ANCOVA specification) in the appendix rather than as a main alternative specification. Further, we could not find a populated pre-analysis plan (Populated PAP) accompanying the manuscript, as recommended by (Banerjee et al., 2020). Although it would appear that the manuscript was published the same time as this recommendation. If the authors had followed this advice, it would have been clear whether the results presented matched the pre-specified analysis or were exploratory analysis. But given the concurrent timing of the publication this is an unreasonable standard for this publication.

4 Conclusion

In this reproduction, we attempt to computationally reproduce and directly replicate the results of Romero et al. (2020). Despite a failed computational reproduction using the publicly provided replication packet in its current form, we successfully make modifications for a revised reproduction of the headline results presented by the original authors. Re-analyzing 798 different specifications to test student learning on English and mathematics, we find that the results are consistently statistically significant but the magnitude of the results are a bit inflated with the preferred specification by the original authors. This provides confidence in the external validity of these findings by policy-makers to utilize public-private partnerships as administrators of publicly funded schools to improve student learning in developing countries. This reproduction contributes

to the Mertonian Norms (Merten, 1942) for the practical application of the open science movement in contemporary scientific research as a validation tool for stakeholders and a corrective tool for researchers.

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5 Figures

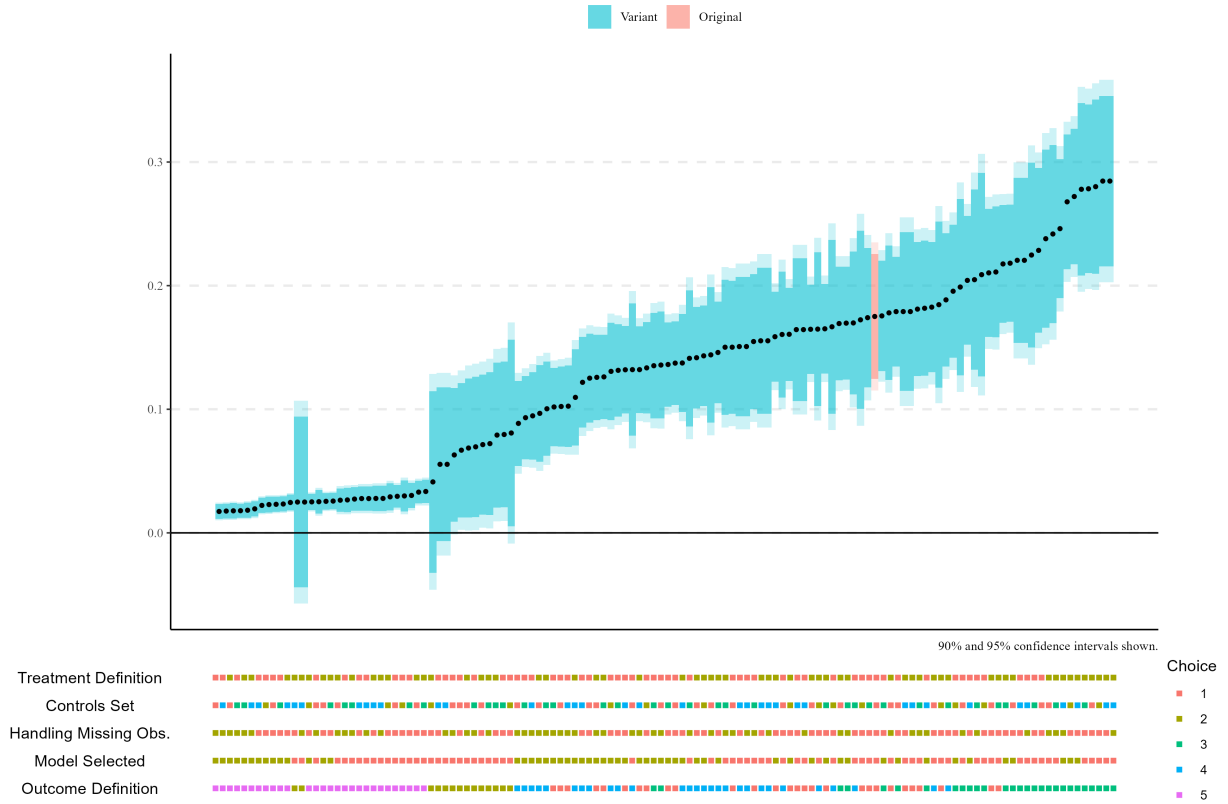


Figure 1: Specification Curve for English Student Test Scores

Notes: This figure presents the multiverse of potential reasonable specifications for estimating the effect of the PSL program on student test scores in English. It is organized using a specification curve (Simonsohn et al., 2020; Klein, 2024). The treatment effect estimates are sorted from smallest to largest with a 95% confidence interval stem from each estimate relative to a null effect of zero treatment effect. The point estimates are in standard deviations. The panel below the curve indicates the combination of decisions culminating into each specific specification. Further details on each specification can be examined in the accompanying Appendix Table A.5.

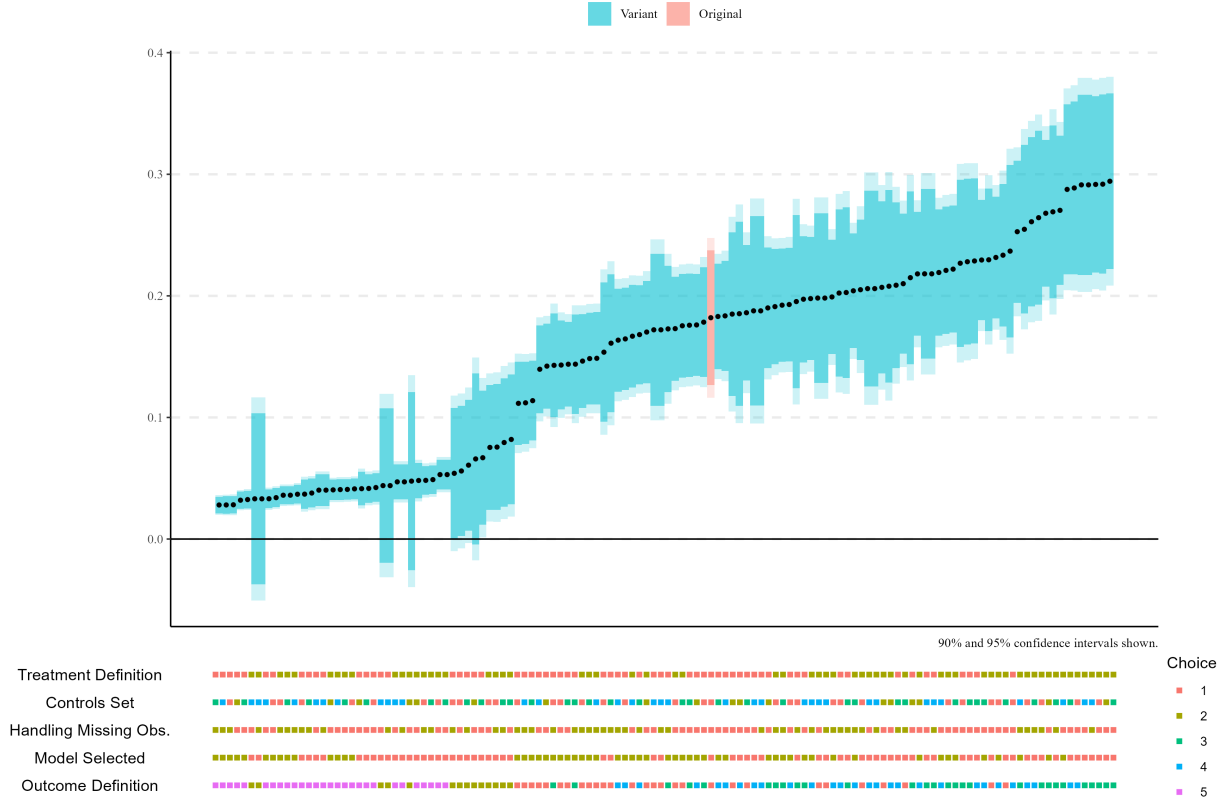


Figure 2: Specification Curve for Mathematics Student Test Scores

Notes: This figure presents the multiverse of potential reasonable specifications for estimating the effect of the PSL program on student test scores in mathematics. It is organized using a specification curve (Simonsohn et al., 2020; Klein, 2024). The treatment effect estimates are sorted from smallest to largest with a 95% confidence interval stem from each estimate relative to a null effect of zero treatment effect. The point estimates are in standard deviations. The panel below the curve indicates the combination of decisions culminating into each specific specification. Further details on each specification can be examined in the accompanying Appendix Table A.6.

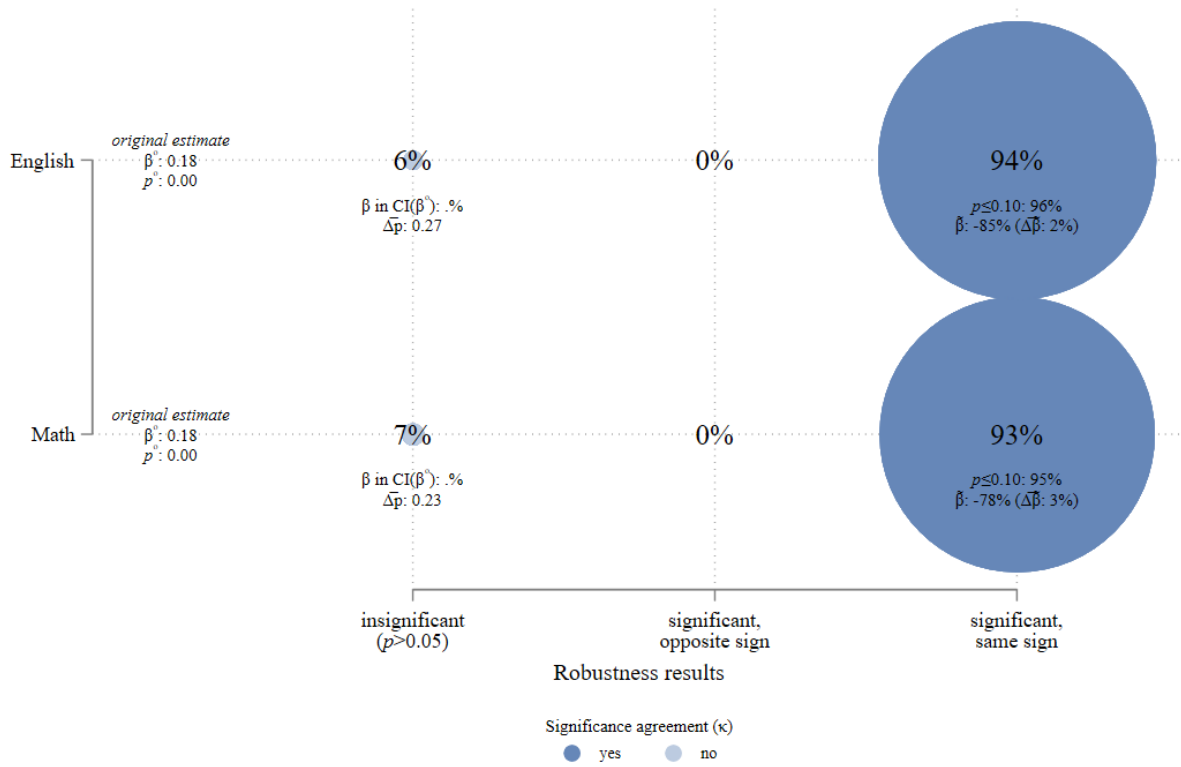


Figure 3: Sensitivity Dashboard of Reproducibility Indicators

Notes: This is a reproducibility sensitivity dashboard developed by through Gunther Bensch's repframe package in Stata. The details on the indicators this summarizes are laid out in detail in Appendix Section A.5. β is the original treatment effect for each outcome of interest. $\tilde{\beta}$ is the median estimated coefficient from the re-analyses measured as % deviation from β . $\Delta \tilde{\beta}$ is the mean relative change of the re-analyses estimates from the reported treatment effect. p is the original reported p-value. Δp is the mean relative change in statistical significance if the p-value is no longer significant.

6 Tables

Table 1: Revised Computational Reproduction of Student Test Scores for English and Mathematics

	First Wave			Second Wave		
	ITT		TOT	ITT		TOT
English Test Scores						
Original	0.09	0.07	0.08	0.17	0.18	0.21
	(0.05)	(0.03)	(0.04)	(0.04)	(0.03)	(0.04)
Reproduced	0.0868	0.0667	0.0680	0.1724	0.1751	0.2103
	(0.0461)	(0.0322)	(0.0357)	(0.0437)	(0.0305)	(0.0313)
Mathematics Test Scores						
Original	0.07	0.05	0.06	0.19	0.18	0.22
	(0.04)	(0.03)	(0.04)	(0.04)	(0.03)	(0.04)
Reproduced	0.0715	0.0526	(0.0681)	0.1850	0.1820	0.2297
	(0.0428)	(0.0313)	(0.0319)	(0.0408)	(0.0335)	(0.0355)

Notes: The rows show the original presented results along with the revised reproduction results for the primary outcomes of interest identified in the pre-analysis plan (Romero et al., 2017) for student test scores in english and mathematics. Each column is a specification presented by the original authors in Table 3 of Romero et al. (2020).

Table 2: Summary of Replication Packet and Computational Reproduction

	Fully	Partially	No
Raw data provided		✓	
Cleaning code provided	✓		
Analysis data provided	✓		
Analysis code provided	✓		
Reproducible from raw data		✓	
Reproducible from analysis data		✓	

Notes: This presentation follows the Robustness Reproducibility in Economics (R2E) Reproducibility Assessment Protocol for reporting computational reproductions: https://www.meta-rep.uni-muenchen.de/projects/project_peters/index.html. Further details on the protocol can be found in Appendix Section A.

Table 3: Summary of Reproducibility Indicators

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Original β	p-value of	Significance	Relative	Relative	Effect Size	Significance	
Outcome	Estimate	Original	Agreement	Effect Size	Significance	Variation	Variation
		Estimate	(RF.1)	(RF.2)	(RF.3)	(RF.4)	(RF.5)
English Test Scores	0.1751	0.0000000	0.9440	0.1434	0.8822	0.1512	1.5402
Math Test Scores	0.1820	0.0000002	0.9280	0.2185	1.0419	0.2107	1.7275
All Outcomes		0.0000001	0.9360	0.1810	0.9620	0.1810	1.6338

Notes: This table shows the reproducibility indicators as elaborated upon in Appendix Section A.5. Columns 2 and 3 show the original reported β coefficients and p-values for the two outcomes of interest in Romero et al. (2020). Column 4 – 8 show the following reproducibility indicators: (RF.1) The statistical significance indicator measures for each outcome the share of the reanalyses that are reported as statistically significant in both the original study and the robustness analysis. (RF.2) The relative effect size indicator measures the mean of the coefficients of all reanalyses for each outcome divided by the original coefficient. (RF.3) The relative significance indicator measures for each outcome the mean of the t/z-values of all the reanalyses divided by the t/z-value of the original result. (RF.4) The effect size variation indicator measures for each outcome the standard deviation of all reanalysis coefficients divided by the standard error of the original coefficient. (RF.5) The significance variation indicator measures the standard deviation of t/z-values of all the robustness analysis paths for each outcome.

Appendix

A Reproducibility Assessment Protocol

The reproducibility assessment protocol for the Robustness and Replicability in Economics (RE2) project standardizes the procedures for evaluating the validity of claims made in the assessed manuscript, document errors and issues with computational reproduction, and provides a template for pre-registering a pre-analysis plan of re-analyses performed in the robustness reproduction process. Below are the main features of this protocol.

A.1 Pure Computational Reproduction

Using the data and scripts provided through the replication packet of the original paper, run the code without correction or changes. You may change the working directory and install missing packages to make the code run. Only consider the results from the original paper which are reported in the tables. Check if the reproduced results match those presented in the paper. Document the errors that prevent the code from running but do not resolve them. Errors which have been identified and can be fixed with be resolved through the revised reproduction. Document the computational reproduction process in a table format – as expressed in Table 2. Note that the different components may achieve degrees of reproducibility, as explained below.

- Reproducible from Raw Data:
 - Mark as "Fully" if both raw data and cleaning code are provided, and utilizing these inputs leads to identical results as in the paper.
 - Mark as "Partially" if raw data and cleaning code are provided, but utilizing these inputs leads to slight deviations in the results.
 - Mark as "No" if no raw data is provided, if raw data is provided without cleaning code, or the code fails to run.
- Reproducible from Analysis Data:
 - Mark as "Fully" if the analysis data and analysis code are provided, and the reproduced results are identical to the paper's results.
 - Mark as "Partially" if the analysis data and analysis code are provided, but there is slight deviation in the reproduced results from the paper.

A.2 Scoping for the Main Outcomes of Interest

Read the original paper. Identify the main research question, the main outcomes of interest, and the tables or figures where the main results for these outcomes of interest are presented. There may be a maximum of five main outcomes of interest. The main outcomes of the paper are defined as those that are mentioned most prominently in the paper (Both statistically significant and null-results qualify) – prominently those mentioned in the Abstract and Introduction. The central

policy recommendations of the paper should derive from these outcomes. Given the subjectivity in their selection, replicators should provide a brief explanation to justify their selection. These main outcomes are the results which will be subject to robustness reproducibility.

The main outcome measures of Romero et al. (2020) are found in Table 3 of the manuscript related to the ITT effects of the PSL program on English and mathematics student test scores. This information is summarized in Table A.1.

Table A.1: Main Outcomes in Romero et al. (2020)

	(1)	(2)
	English Student Test Score	Mathematics Student Test Score
Location in Original Paper	Table 3, Column 5	Table 3, Column 5
Estimate	0.18	0.18
Standard Error	0.03	0.03
Standard Error type	Clustered at school level	Clustered at school level
p-value	< 0.001	< 0.001
95-Confidence Interval	Not Reported	Not Reported
Level of analysis	Individual (student)	Individual (student)
Type of variable	Continuous	Continuous
Units	Standard deviation	Standard deviation
Number of observations	3492	3492
Outcome pre-specified	Yes	Yes
Outcome construction pre-specified	Yes	Yes

A.3 Revised Reproduction

Regardless of a successful or failed computational reproduction of the main tables in the original paper, the revised reproduction aims to identify potential objective coding errors. Objective errors are defined as those that either lead to incorrect reproduced results or the coding runs contrary to what is reported by the authors in the manuscript. For example, this may be misspecification of a variable as described in the manuscript, or adding a dummy for missing control variables not mentioned in the manuscript. These objective errors contrast with the reasonable analytical decisions which are challenged in the robustness reproduction. Replicators correct these identified errors and record their impact on the reproduced results. For efficiency, focus on four critical coding sections for review:

- Main Outcomes: Sections related to generating main outcomes.
- Treatment Variables: Sections that handles the treatment variable definitions.
- Control Variables: Sections relevant to control variables used in the pertinent estimations.
- Estimation Process: Sections implementing statistical models or methods.

In particular, replicators are to be vigilant of the following objective coding errors:

- Variable misspecification
- Coding switcheroos (e.g., variable accidentally coded different than claimed by the authors)

- Outlier management
- Missing data handling
- Sensitive commands for cleaning data (e.g., merging, appending, reshaping, dropping data)

A.4 Robustness Reproduction

Replicators consider a series of potential robustness tests to the analytical decisions made by the authors within the framework of a multiverse of potential outcomes (Steege et al., 2016; Simonsohn et al., 2020). Further, the replicators are to estimate the combination of all potential reasonable decisions that lead to different ‘analysis paths’. There is no limit to the number of potential robustness checks to test. Before proceeding with any proposed robustness check, replicators are to provide a brief justification why this is a sensible alternative decision the research may have made.

Replicators are to compile a comprehensive list of the specific checks they intend to implement for each main outcome and formally register them prior to analysis. Replicators pre-register all analytical decisions and the potential choices related to them alongside the original analytical decision of the authors in a pre-analysis plan uploaded to the Open Science Platform (OSF). Replicators have the freedom to include further robustness checks not pre-registered, but should be acknowledged that these are non-pre-registered for transparency.

Of particular interest are the role of control variables and whether they operate as a collider (bad control) or a confounder (good control) in minimizing omitted variable bias. The suggested dimensions for robustness checks include (but are not limited to):

- Definition of the analysis sample
- Definition of the outcome variables
- Definition of the treatment variables
- Inclusion and definition of control variables
- Outlier management
- Missing observations
- Model selection

For each set of analytical decisions related to the dimensions for robustness checks, replicators should provide a brief description and motivation for their use. The multiverse robustness tests should present the estimated effect size, standard error, t/z-value and p-value for each re-analysis.

A.5 Reproducibility Indicators

Replicators should report a table for each main outcome with the results of the robustness tests. This table should include the following reproducibility indicators. For this reproduction, the re-analysis results are presented in Appendix Table A.5 for students’ English test scores and Appendix Table A.6 for students’ mathematics test scores. These are summarized by the specification curves¹

¹This specification is a modification of Simonsohn et al. (2020) adapted by Nick Huntington-Klien in the `r2especcurve` package: <https://github.com/NickCH-K/r2especcurve>

presented by Figures 1 and 2. Whereas the reproducibility indicators are presented in Figure ??.

Below are is the protocol for reporting for the original results reported as statistically significant (e.g., as is the case for Romero et al. (2020)). This information is suggested to be presented in a Sensitivity Dashboard.

1. Statistical Significance: % of robustness tests analysis paths in the same direction as the original effect and a two-sided p-value <0.05 .
 - The original analysis path should not be included among the analysis paths in the estimation of the statistical significance indicator.
2. Relative Effect Size: Mean effect size of all the robustness tests analysis paths divided by the original effect size.
 - The original analysis path should not be included among the analysis paths in the estimation of the mean effect size of all the robustness tests analysis paths.
 - Only the robustness tests analysis paths that use the same effect size units as the original result should be included. This excludes, for example, robustness tests analysis paths with effect sizes in logs, while the original results are in levels.
3. Relative t/z-value: Mean t/z-value of all the robustness tests analysis paths divided by the t/z-value of the original effect.
 - The original analysis path should not be included among the analysis paths in the estimation of the mean t/z-value of all the robustness tests analysis paths.
4. Variation indicator: effect sizes: The standard deviation in effect sizes of all the robustness tests analysis paths divided by the standard error of the original effect size.
 - This indicator should not be estimated for “main outcomes” involving two coefficients.
 - If the original analysis is included as one analysis path in the multiverse robustness test that analysis path should be included in the estimation of the standard deviation.
 - Only the robustness tests analysis paths that use the same effect size units as the original result should be included in the estimation of the standard deviation. This excludes, for example, robustness tests analysis paths with effect sizes in logs, while the original results are in levels.
5. Variation indicator: t/z-values: Standard deviation of t/z-value of all the robustness tests analysis paths.
 - If the original analysis is included as one analysis path in the multiverse robustness test, that analysis path should be included in the estimation of the standard deviation.

A.6 Subjective Assessment of Replicability

Upon completion of the reproduction, replicators are to subjectively evaluate seven items and provide a rating on a 0 – 10 scale. Below we provide a rating for each of these items as well as instructions for assessment and justification for the decision.

1. Robustness Reproducibility: 10/10

- Instructions: After carefully reflecting on your robustness reproducibility results, please rate the overall robustness reproducibility of the paper you have tested the robustness reproducibility of, using a 0-10 scale ranging from “not at all robust results” (0) to “very robust results” (10).
- Justification: The main results are robust to a variety of changes to the analytical decisions by the original authors regarding sample for analysis, treatment definition, handling missed observations, including different controls, and re-defining the outcome measures.

2. Identification Strategy: 10/10

- Instructions: Based on your analysis, please rate the strength of the identification of the paper you have tested the robustness reproducibility of, using a 0-10 scale ranging from “very weak identification” (0) to “very strong identification” (10).
- Justification: The identification used a matched pair RCT following a pre-treatment sample of students for outcome measures. This overcomes the omitted variable bias from student selection on treatment status from school-level RCT, randomized lotteries, or natural experimental expansion of charter schools.

3. Pre-analysis Plan (PAP) Quality: 10/10

- Instructions: Rate the PAP in terms of its quality on a 0-10 scale from “very uninformative pre-analysis plan” (0) to “very informative pre-analysis plan” (10).
- Justification: Very informative. Expounds on empirical methodology, intervention implementation by partners, baseline characteristics, an intervention timeline, clearly stated hypotheses, and defined outcome measures mapped to the survey instrument categorized by families of outcomes. There is a pre-specified model specification using linear regression, and consideration of heterogeneous treatment effects.

4. Adherence to PAP: 9/10

- Instructions: Rate to what extent the PAP was followed on a 0-10 scale from “not followed at all” (0) to “completely followed” (10).
- Justification: Constructed the outcome measures as described. Use the pre-specified models and pre-specified controls. A small deviation in that they only report equation 3 (i.e., the ANCOVA) in the appendix. They do report some identified ‘main outcomes’ as secondary outcomes in the final manuscript: attendance, enrollment, and selection.²

5. Original Paper’s Importance: 9/10

²We could not find a populated pre-analysis plan (Populated PAP) accompanying the manuscript, as recommended by (Banerjee et al., 2020). Although it would appear that the manuscript was published the same time as this recommendation. If the authors had followed this advice, it would have been clear whether the results presented matched the pre-specified analysis or were exploratory analysis.

- Instructions: How important are the results/insights from this study, using a 0-10 scale ranging from “not important at all” (0) to “extremely important” (10).
- Justification: This shows an important result that public-private partnerships have value for school settings and the value of outsourcing school management.

6. Original Paper’s Generalizability (External Validity): 6/10

- Instructions: In how far are the results generalizable, using a 0-10 scale ranging from “not at all generalizable” (0) to “very generalizable” (10).
- Justification: The generalizability of the results are limited the the unique structure of the Liberian Ministry of Education. The Liberian government funds their schools via mostly aid donations – some of whom participated as providers in this study – as compared to a tax as many other developing countries. Therefore, the ability to expand public-private partnerships in other settings depends heavily on (1) the financing structure that allows for outsourcing school management, and (2) that there is adequate provisions in the market for partnerships with private providers to manage schools. This is to say, the providers in this setting are mainly non-profit organizations. Would the results hold with a different set of incentives with primarily for-profit or religious institutions managing outsourced schools?

7. Original Paper’s Construct Validity: 8/10

- Instructions: To what extent is the information provided sufficient to assess the construct validity? Use a 0-10 scale ranging from “not at all sufficient” (0) to “completely sufficient” (10).
- Justification: This study uses item response theory (IRT) to construct on school measurement from independent assessments of student learning (i.e., test scores). Additionally, they collect admin data as supplementary information. The test scores are defined in a standard fashion consistent with the education literature.

B Identified Errors in Computational Reproduction

This section documents that errors encountered by the replicator in computationally reproducing the results of Romero et al. (2020) using their publicly provided replication packet Romero et al. (2018). The section provides details that support the failed computational reproduction reported in Section 2.1. There are two types of coding errors that were encountered. The first are errors by design from the authors of replication packet who restricted data but provided scripts to (in principal) show how a reproduction of the results could be performed. The second are unexpected errors that broke the provided scripts given the provided data. Despite a diligent effort to resolve these unintentional errors, it lead to a failed computational reproduction. This resulted in a modified revised reproduction focusing on solely reproducing the main results presented in Table 3 of Romero et al. (2020). This revised reproduction is reported in Section 2.2.

B.1 Errors by Design

Below are a list of scripts that are unable to be run. These are accompanied by the reason this script cannot be run as well as the implication for main manuscript results (i.e., not supplementary appendix results) which cannot be computationally reproduced. Three important acronyms used and a pseudonym from providers are listed below.

- PII: Personally identifiable information (i.e., student, teacher, household, or provider information)
- EMIS: Education Management Information System (i.e., administrative data from the Ministry of Education – Liberia)
- HEIS: Household Income and Expenditure Survey 2014-2015 (Obtainable from <http://microdata.worldbank.org/index.php/catalog/2563>.)
- Operators: The implementing providers for the PSL treatment arms (i.e., the private institutions running government schools)

Table A.2: Errors by Design Preventing Computational Reproduction

Coding Script	Reason	Irreproducible Results
01_PrepAdmin_Data.do	Restricted access to EMIS data	
02_ReadTreatmentFile.do	Restricted access to PII data	
02b_AnonimizeData.do	Restricted access to PII data	
03_BalanceEMIS_n_ExtValidity.do	Restricted access to EMIS data	
9a_TeacherCharact.do	Restricted access to EMIS data	
13a_OperatorComparisons.do	Restricted access to Operator data	Table 10
13b_SchoolsOperators.do	Restricted access to PII data	
15_HEIS.do	Restricted access to HEIS data	
16_budgets.do	Restricted access to Operator data	Figure 2
17_SomeStats.do	Restricted access to Operator data	
18_SimulateEffectsWithoutSome.do	Restricted access to Operator data	
01_DistanceToPavedRoad.R	Restricted access to PII & EMIS data	
02_MakeMaps_Paper.R	Restricted access to PII data	Figure 3
04_GraphHIES_Data.R	Restricted access to HEIS data	
05a_heatmap.R	Restricted access to Operator data	Figure 1
06a_Bayesian_Learning.R	Restricted access to Operator data	
07a_Bayesian_Learning_ToT.R	Restricted access to Operator data	
08a_Bayesian_Other.R	Restricted access to Operator data	Table 10
09a_Bayesian_Learning_Plot.R	Restricted access to Operator data	
10_Simulations.R	Restricted access to Operator data	
11_Budgets.R	Restricted access to Operator data	Figure 2

B.2 Unexpected Errors

Below are a list of unexpected errors identified during the computational reproduction. These errors are identified by the script they pertain to and the line of code they appear upon, along with the error message that was provided.

Given the errors below, we am unable to computationally reproduce the following tables for the main manuscript: Table 2, Table 3, Table 4, Table 5, Table 6, Table 7, Table 8, and Table 9. All of these tables are estimated and constructed within 12a_MidlineResults.do, and the failure to successfully run this script results in irreproducibility of these tables.

Table A.3: Unexpected Errors Preventing Computational Reproduction

	Coding Script	Line	Error Message
1.	00_MasterFile.do	57	'exit' command appears to prevent TvsC_tables_ML.do from running properly.
2.	06_Student_BalanceSampling.do	53	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado)
3.	06_Student_BalanceSampling.do	78	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
4.	07b_Student_Prep_Midline.do	52	function cleanup_before_saving() not declared in c lass Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
5.	07b_Student_Prep_Midline.do	53	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
6.	07b_Student_Prep_Midline.do	54	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
7.	10_CombineDatasets.do	54	file C:/Users/original_data/PSL Endline Dataverse Files// Analysis/usedata_public/TeacherAttendance.dta cannot be modified or erased; likely cause is read-only directory or file r(608);
8.	11_Baseline_Balance.do	27	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
9.	11_Baseline_Balance.do	37	function cleanup_before_saving() not declared in ; class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000)
10.	11_Baseline_Balance.do	68	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
11.	11_Baseline_Balance.do	81	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
12.	12a_MidlineResults.do	20	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
13.	12a_MidlineResults.do	35	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);

14.	12a_MidlineResults.do	84	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
15.	12a_MidlineResults.do	84	parentheses unbalanced r(132)
16.	12a_MidlineResults.do	88	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
17.	12a_MidlineResults.do	180	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
18.	12a_MidlineResults.do	238	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
19.	12a_MidlineResults.do	246	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
20.	12a_MidlineResults.do	253	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
21.	12a_MidlineResults.do	258	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
22.	12a_MidlineResults.do	330	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
23.	12a_MidlineResults.do	388	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
24.	12a_MidlineResults.do	429	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
25.	12a_MidlineResults.do	433	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
26.	12a_MidlineResults.do	444	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
27.	12a_MidlineResults.do	448	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
28.	12a_MidlineResults.do	682	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);

29.	12a_MidlineResults.do	691	function cleanup_before_saving() not declared in
			class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
30.	12a_MidlineResults.do	700	function cleanup_before_saving() not declared in
			class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
31.	12a_MidlineResults.do	709	function cleanup_before_saving() not declared in
			class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
32.	12a_MidlineResults.do	718	function cleanup_before_saving() not declared in
			class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
33.	12a_MidlineResults.do	755	function cleanup_before_saving() not declared in
			class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
34.	12a_MidlineResults.do	761	function cleanup_before_saving() not declared in
			class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
35.	12a_MidlineResults.do	762	function cleanup_before_saving() not declared in
			class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
36.	12a_MidlineResults.do	850	function cleanup_before_saving() not declared in
			class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
37.	12a_MidlineResults.do	858	function cleanup_before_saving() not declared in
			class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
38.	12a_MidlineResults.do	914	function cleanup_before_saving() not declared in
			class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
39.	12a_MidlineResults.do	922	function cleanup_before_saving() not declared in
			class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
40.	12a_MidlineResults.do	963	function cleanup_before_saving() not declared in
			class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
41.	12a_MidlineResults.do	1361	function cleanup_before_saving() not declared in
			class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
42.	12a_MidlineResults.do	1370	function cleanup_before_saving() not declared in
			class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
43.	12a_MidlineResults.do	1380	function cleanup_before_saving() not declared in

44.	12a_MidlineResults.do	1387	class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000); function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
45.	12a_MidlineResults.do	1397	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
46.	12a_MidlineResults.do	1405	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
47.	12a_MidlineResults.do	1501	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
48.	12a_MidlineResults.do	1512	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
49.	12a_MidlineResults.do	1524	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
50.	12a_MidlineResults.do	1534	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
51.	12a_MidlineResults.do	1545	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
52.	12a_MidlineResults.do	1555	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
53.	12a_MidlineResults.do	1567	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
54.	12a_MidlineResults.do	1578	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
55.	12a_MidlineResults.do	1603	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
56.	12a_MidlineResults.do	1655	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
57.	12a_MidlineResults.do	1664	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);

58.	12a_MidlineResults.do	1676	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
59.	12a_MidlineResults.do	1721	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
60.	12a_MidlineResults.do	1729	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
61.	12a_MidlineResults.do	1768	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
62.	12a_MidlineResults.do	1778	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
63.	12a_MidlineResults.do	1786	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
64.	12a_MidlineResults.do	1796	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
65.	12a_MidlineResults.do	1802	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
66.	12a_MidlineResults.do	1812	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
67.	12a_MidlineResults.do	1819	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
68.	12a_MidlineResults.do	1829	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
69.	12a_MidlineResults.do	1835	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
70.	12a_MidlineResults.do	1871	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
71.	12a_MidlineResults.do	1960	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
72.	12a_MidlineResults.do	1971	function cleanup_before_saving() not declared in

73.	12a_MidlineResults.do	1978	class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000); function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
74.	12a_MidlineResults.do	1988	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
75.	12a_MidlineResults.do	2005	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
76.	12a_MidlineResults.do	2012	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
77.	12a_MidlineResults.do	2063	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
78.	12a_MidlineResults.do	2073	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
79.	12a_MidlineResults.do	2081	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
80.	12a_MidlineResults.do	2270	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
81.	12a_MidlineResults.do	2283	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
82.	12a_MidlineResults.do	2292	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
83.	12a_MidlineResults.do	2302	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
84.	12a_MidlineResults.do	2308	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
85.	12a_MidlineResults.do	2370	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
86.	12a_MidlineResults.do	2380	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);

87.	12a_MidlineResults.do	2395	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
88.	12a_MidlineResults.do	2405	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
89.	12a_MidlineResults.do	2411	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
90.	12a_MidlineResults.do	2482	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
91.	12a_MidlineResults.do	2501	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
92.	12a_MidlineResults.do	2556	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
93.	12a_MidlineResults.do	2565	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
94.	12a_MidlineResults.do	2570	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
95.	12a_MidlineResults.do	2580	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
96.	12a_MidlineResults.do	2594	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
97.	12a_MidlineResults.do	2603	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
98.	12a_MidlineResults.do	2609	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
99.	12a_MidlineResults.do	2680	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
100.	12a_MidlineResults.do	2692	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
101.	12a_MidlineResults.do	2697	function cleanup_before_saving() not declared in

102.	12a_MidlineResults.do	2702	class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000); function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
103.	12a_MidlineResults.do	2711	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
104.	12a_MidlineResults.do	2717	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
105.	12a_MidlineResults.do	2726	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
106.	12a_MidlineResults.do	2738	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
107.	12a_MidlineResults.do	2747	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
108.	12a_MidlineResults.do	2756	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
109.	12a_MidlineResults.do	2848	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
110.	12a_MidlineResults.do	2857	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
111.	12a_MidlineResults.do	2862	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
112.	12a_MidlineResults.do	2880	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
113.	12a_MidlineResults.do	2889	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
114.	12a_MidlineResults.do	2894	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
115.	12a_MidlineResults.do	2954	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);

116.	12a_MidlineResults.do	2963	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
117.	12a_MidlineResults.do	2972	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
118.	12a_MidlineResults.do	3051	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
119.	12a_MidlineResults.do	3060	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
120.	12a_MidlineResults.do	3066	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
121.	13c_OperatorComparisons_ML_RI.do	22	"file C:/Users/PSL Endline Dataverse Files//Analysis/usedata/ school_info_crosswalk.dta not found
122.	14_TimeTable.do	24	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
123.	14_TimeTable.do	25	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
124.	14_TimeTable.do	26	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
125.	14_TimeTable.do	27	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
126.	14_TimeTable.do	28	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);
127.	14_TimeTable.do	29	function cleanup_before_saving() not declared in class Factor (57 lines skipped) (error occurred while loading reghdfe.ado) r(3000);

C Supplementary Material for Specification Curves

C.1 Summary of Robustness Checks by Analytical Decision

The re-analysis (robustness checks) are performed within the multiverse framework Simmons et al. (2021) that the presented results in a manuscript are but one analysis path in a multiverse of alternative paths determined by the researchers choice over analytical decisions. This section reports the analytical dimensions and the various choices within them that we re-analyze in this replication study. These choices are pre-registered in a pre-analysis plan on the Open Science Platform which can be found here: <https://osf.io/hxsuz>. Table A.4 reports the identifier for each analytical choice along with a justification for it's selection. Some of these choices are infeasible to estimate. E.1.2 and M.1.2 is infeasible to estimate because provider assignment to students is restricted data not provided by the replication packet. Additionally, some combinations of these analytical choices are not reasonable. M/E.3.2 for no controls and M/E.4.2 for not replacing missing controls doesn't make senses because there are no control variables to modify. M/E.5.2 for an ANCOVA cannot pair with M/E.6.2 for using baseline values as the outcome measure. This makes the outcome perfectly correlated with a control and forces the treatment effect to be omitted.

Table A.4: Alternative Choices for Analytical Decisions

English Label	Math Label	Choice	Rational
Dimension: Treatment Definition			
E.2.1	M.2.1	Intention-to-Treat	Original choice
E.2.2	M.2.2	Treatment-on-Treated	Non-compliance by providers
Dimension: Control Variables			
E.3.1	M.3.1	Pre-specified set of controls	Original choice
E.3.2	M.3.2	No controls	Results uninfluenced by other factors
E.3.3	M.3.3	Only student-level controls	Sensitivity by student factors
E.3.4	M.3.4	Only school-level controls	Sensitivity to school environment
Dimension: Handling Missing Observations			
E.4.1	M.4.1	Replace missing controls with dummy indicator	Original choice
E.4.2	M.4.2	Do not replace missing control values	Correction may be unnecessary
Dimension: Model Selection			
E.5.1	M.5.1	RCT	Original choice
E.5.2	M.5.2	ANCOVA	Control for baseline outcome measures
Dimension: Defining Outcomes			
E.6.1	M.6.1	IRT standardized scores	Original choice
E.6.2	M.6.2	Baseline outcome value	Captures initial treatment from providers
E.6.3	M.6.3	Standardized scores by grade level	Removes variance across grades
E.6.4	M.6.4	PCA predicting test performance	Captures important factors indicative of learning
E.6.5	M.6.5	% of correct answers on test	Direct measure of test performance

Notes: This shows the alternative choices for each dimension of analytic decision in specifying a model. The combination of these rows (choices) culminate into the combinations that are estimated for re-analysis. Due to data restrictions E.1.2 and M.1.2 were not implemented as part of these combinations.

C.2 Detailed Re-analysis Results Presented in Specification Curves

This section provides table results for the specification curves provided in the manuscript. Given the necessity to simplify information in the presentation of the specification curves, it may be difficult to identify the combinations or statistics resulting from a specific combination. The two tables below are the table analogs to the specification curves sorting combinations in the same order as they are presented in the specification curves.

Table A.5: Re-analyses Performed in the Specification Curve for Figure 1

(1) Combination Label	(2) Dimension Sample	(3) Dimension Treatment	(4) Dimension Controls	(5) Dimension Missing Obs.	(6) Dimension Model	(7) Dimension Outcome	(8) Treatment Effect	(9) SE	(10) t-value	(11) p-value	(12) Obs.
e457	E.1.1	E.2.1	E.3.1	E.4.2	E.5.2	E.6.5	0.017322	0.003576	4.843649	2.77E-06	3291
e475	E.1.1	E.2.1	E.3.4	E.4.2	E.5.2	E.6.5	0.017651	0.003602	4.900724	2.15E-06	3356
e460	E.1.1	E.2.2	E.3.1	E.4.2	E.5.2	E.6.5	0.017835	0.00383	4.656352	6.3E-06	3291
e469	E.1.1	E.2.1	E.3.3	E.4.2	E.5.2	E.6.5	0.017934	0.003511	5.108531	8.35E-07	3291
e472	E.1.1	E.2.2	E.3.3	E.4.2	E.5.2	E.6.5	0.018309	0.003739	4.896784	2.19E-06	3291
e478	E.1.1	E.2.2	E.3.4	E.4.2	E.5.2	E.6.5	0.019429	0.003772	5.150772	6.87E-07	3356
e451	E.1.1	E.2.1	E.3.4	E.4.1	E.5.2	E.6.5	0.022219	0.003711	5.98801	1.09E-08	3492
e439	E.1.1	E.2.1	E.3.2	E.4.1	E.5.2	E.6.5	0.022907	0.003833	5.975707	1.16E-08	3492
e433	E.1.1	E.2.1	E.3.1	E.4.1	E.5.2	E.6.5	0.023105	0.003683	6.273832	2.46E-09	3492
e445	E.1.1	E.2.1	E.3.3	E.4.1	E.5.2	E.6.5	0.023351	0.003712	6.290292	2.25E-09	3492
e454	E.1.1	E.2.2	E.3.4	E.4.1	E.5.2	E.6.5	0.024584	0.003901	6.301624	2.12E-09	3492
e118	E.1.1	E.2.2	E.3.4	E.4.1	E.5.1	E.6.2	0.024939	0.041829	0.596206	0.551799	3356
e142	E.1.1	E.2.2	E.3.4	E.4.2	E.5.1	E.6.2	0.024939	0.041829	0.596206	0.551799	3356
e442	E.1.1	E.2.2	E.3.2	E.4.1	E.5.2	E.6.5	0.025082	0.003953	6.344244	1.69E-09	3492
e409	E.1.1	E.2.1	E.3.1	E.4.2	E.5.1	E.6.5	0.025213	0.005747	4.38713	1.97E-05	3291
e436	E.1.1	E.2.2	E.3.1	E.4.1	E.5.2	E.6.5	0.025503	0.004047	6.302249	2.11E-09	3492
e448	E.1.1	E.2.2	E.3.3	E.4.1	E.5.2	E.6.5	0.025719	0.004009	6.415584	1.15E-09	3492
e412	E.1.1	E.2.2	E.3.1	E.4.2	E.5.1	E.6.5	0.026423	0.005779	4.572432	9.03E-06	3291
e421	E.1.1	E.2.1	E.3.3	E.4.2	E.5.1	E.6.5	0.02665	0.005937	4.489001	1.29E-05	3291
e424	E.1.1	E.2.2	E.3.3	E.4.2	E.5.1	E.6.5	0.02735	0.005812	4.705825	5.08E-06	3291
e403	E.1.1	E.2.1	E.3.4	E.4.1	E.5.1	E.6.5	0.027764	0.006016	4.615234	7.35E-06	3492
e427	E.1.1	E.2.1	E.3.4	E.4.2	E.5.1	E.6.5	0.027764	0.006016	4.615234	7.35E-06	3492
e406	E.1.1	E.2.2	E.3.4	E.4.1	E.5.1	E.6.5	0.027858	0.006227	4.47349	1.34E-05	3492
e430	E.1.1	E.2.2	E.3.4	E.4.2	E.5.1	E.6.5	0.027858	0.006227	4.47349	1.34E-05	3492

e394	E.1.1	E.2.2	E.3.2	E.4.1	E.5.1	E.6.5	0.029149	0.006604	4.413872	1.73E-05	3492
e385	E.1.1	E.2.1	E.3.1	E.4.1	E.5.1	E.6.5	0.029569	0.005477	5.399232	2.05E-07	3492
e391	E.1.1	E.2.1	E.3.2	E.4.1	E.5.1	E.6.5	0.029876	0.007534	3.965598	0.000105	3492
e397	E.1.1	E.2.1	E.3.3	E.4.1	E.5.1	E.6.5	0.03027	0.006156	4.916801	1.94E-06	3492
e388	E.1.1	E.2.2	E.3.1	E.4.1	E.5.1	E.6.5	0.032967	0.00558	5.908086	1.64E-08	3492
e400	E.1.1	E.2.2	E.3.3	E.4.1	E.5.1	E.6.5	0.033477	0.005716	5.856818	2.13E-08	3492
e106	E.1.1	E.2.2	E.3.2	E.4.1	E.5.1	E.6.2	0.041222	0.044499	0.926343	0.355529	3356
e115	E.1.1	E.2.1	E.3.4	E.4.1	E.5.1	E.6.2	0.05546	0.037676	1.472044	0.142785	3356
e139	E.1.1	E.2.1	E.3.4	E.4.2	E.5.1	E.6.2	0.05546	0.037676	1.472044	0.142785	3356
e121	E.1.1	E.2.1	E.3.1	E.4.2	E.5.1	E.6.2	0.063038	0.032788	1.922574	0.056139	3291
e97	E.1.1	E.2.1	E.3.1	E.4.1	E.5.1	E.6.2	0.066893	0.032901	2.033125	0.043533	3356
e124	E.1.1	E.2.2	E.3.1	E.4.2	E.5.1	E.6.2	0.068643	0.0341	2.012973	0.045633	3291
e133	E.1.1	E.2.1	E.3.3	E.4.2	E.5.1	E.6.2	0.069603	0.034377	2.024683	0.044403	3291
e100	E.1.1	E.2.2	E.3.1	E.4.1	E.5.1	E.6.2	0.071398	0.034252	2.084502	0.03855	3356
e136	E.1.1	E.2.2	E.3.3	E.4.2	E.5.1	E.6.2	0.072205	0.035018	2.061904	0.040678	3291
e112	E.1.1	E.2.2	E.3.3	E.4.1	E.5.1	E.6.2	0.079105	0.035568	2.224033	0.027411	3356
e109	E.1.1	E.2.1	E.3.3	E.4.1	E.5.1	E.6.2	0.079573	0.035733	2.226856	0.027218	3356
e103	E.1.1	E.2.1	E.3.2	E.4.1	E.5.1	E.6.2	0.080798	0.045674	1.769011	0.078614	3356
e361	E.1.1	E.2.1	E.3.1	E.4.2	E.5.2	E.6.4	0.08856	0.020829	4.251726	3.43E-05	3291
e373	E.1.1	E.2.1	E.3.3	E.4.2	E.5.2	E.6.4	0.093179	0.020433	4.560174	9.51E-06	3291
e379	E.1.1	E.2.1	E.3.4	E.4.2	E.5.2	E.6.4	0.094672	0.0214	4.423901	1.69E-05	3356
e364	E.1.1	E.2.2	E.3.1	E.4.2	E.5.2	E.6.4	0.096691	0.023718	4.076742	6.89E-05	3291
e376	E.1.1	E.2.2	E.3.3	E.4.2	E.5.2	E.6.4	0.10039	0.023106	4.344692	2.35E-05	3291
e85	E.1.1	E.2.1	E.3.3	E.4.2	E.5.2	E.6.1	0.101873	0.01921	5.303029	3.37E-07	3291
e73	E.1.1	E.2.1	E.3.1	E.4.2	E.5.2	E.6.1	0.102242	0.019621	5.210911	5.19E-07	3291
e91	E.1.1	E.2.1	E.3.4	E.4.2	E.5.2	E.6.1	0.102502	0.020041	5.114726	8.11E-07	3356
e382	E.1.1	E.2.2	E.3.4	E.4.2	E.5.2	E.6.4	0.109609	0.023622	4.640118	6.75E-06	3356
e355	E.1.1	E.2.1	E.3.4	E.4.1	E.5.2	E.6.4	0.121792	0.022227	5.479476	1.39E-07	3492
e337	E.1.1	E.2.1	E.3.1	E.4.1	E.5.2	E.6.4	0.125155	0.021848	5.72854	4.07E-08	3492

e76	E.1.1	E.2.2	E.3.1	E.4.2	E.5.2	E.6.1	0.125819	0.020901	6.019633	9.83E-09	3291
e88	E.1.1	E.2.2	E.3.3	E.4.2	E.5.2	E.6.1	0.126192	0.020652	6.110532	6.16E-09	3291
e343	E.1.1	E.2.1	E.3.2	E.4.1	E.5.2	E.6.4	0.130655	0.023872	5.473083	1.43E-07	3492
e349	E.1.1	E.2.1	E.3.3	E.4.1	E.5.2	E.6.4	0.13149	0.022656	5.803699	2.79E-08	3492
e67	E.1.1	E.2.1	E.3.4	E.4.1	E.5.2	E.6.1	0.131951	0.021461	6.148361	4.75E-09	3492
e313	E.1.1	E.2.1	E.3.1	E.4.2	E.5.1	E.6.4	0.132044	0.032432	4.071437	7.04E-05	3291
e94	E.1.1	E.2.2	E.3.4	E.4.2	E.5.2	E.6.1	0.132061	0.021297	6.200946	3.85E-09	3356
e55	E.1.1	E.2.1	E.3.2	E.4.1	E.5.2	E.6.1	0.133483	0.022384	5.963339	1.24E-08	3492
e277	E.1.1	E.2.1	E.3.3	E.4.2	E.5.2	E.6.3	0.135222	0.025811	5.238954	4.55E-07	3291
e265	E.1.1	E.2.1	E.3.1	E.4.2	E.5.2	E.6.3	0.135772	0.026176	5.186796	5.81E-07	3291
e61	E.1.1	E.2.1	E.3.3	E.4.1	E.5.2	E.6.1	0.136214	0.020693	6.582713	4.68E-10	3492
e49	E.1.1	E.2.1	E.3.1	E.4.1	E.5.2	E.6.1	0.137329	0.020523	6.691391	2.58E-10	3492
e358	E.1.1	E.2.2	E.3.4	E.4.1	E.5.2	E.6.4	0.137434	0.024081	5.707098	4.53E-08	3492
e325	E.1.1	E.2.1	E.3.3	E.4.2	E.5.1	E.6.4	0.141201	0.033414	4.225804	3.81E-05	3291
e340	E.1.1	E.2.2	E.3.1	E.4.1	E.5.2	E.6.4	0.141704	0.024904	5.690021	4.93E-08	3492
e346	E.1.1	E.2.2	E.3.2	E.4.1	E.5.2	E.6.4	0.14321	0.024658	5.807775	2.73E-08	3492
e316	E.1.1	E.2.2	E.3.1	E.4.2	E.5.1	E.6.4	0.144009	0.033165	4.342262	2.37E-05	3291
e352	E.1.1	E.2.2	E.3.3	E.4.1	E.5.2	E.6.4	0.145932	0.024936	5.852349	2.18E-08	3492
e328	E.1.1	E.2.2	E.3.3	E.4.2	E.5.1	E.6.4	0.150189	0.033123	4.534252	1.06E-05	3291
e25	E.1.1	E.2.1	E.3.1	E.4.2	E.5.1	E.6.1	0.150213	0.032578	4.610812	7.66E-06	3291
e307	E.1.1	E.2.1	E.3.4	E.4.1	E.5.1	E.6.4	0.150737	0.034198	4.407784	1.77E-05	3492
e331	E.1.1	E.2.1	E.3.4	E.4.2	E.5.1	E.6.4	0.150737	0.034198	4.407784	1.77E-05	3492
e37	E.1.1	E.2.1	E.3.3	E.4.2	E.5.1	E.6.1	0.154782	0.033135	4.671274	5.9E-06	3291
e310	E.1.1	E.2.2	E.3.4	E.4.1	E.5.1	E.6.4	0.155393	0.03579	4.341835	2.33E-05	3492
e334	E.1.1	E.2.2	E.3.4	E.4.2	E.5.1	E.6.4	0.155393	0.03579	4.341835	2.33E-05	3492
e70	E.1.1	E.2.2	E.3.4	E.4.1	E.5.2	E.6.1	0.158684	0.022089	7.18394	1.63E-11	3492
e289	E.1.1	E.2.1	E.3.1	E.4.1	E.5.1	E.6.4	0.160505	0.030873	5.198952	5.3E-07	3492
e58	E.1.1	E.2.2	E.3.2	E.4.1	E.5.2	E.6.1	0.160656	0.022825	7.038441	3.73E-11	3492
e19	E.1.1	E.2.1	E.3.4	E.4.1	E.5.1	E.6.1	0.16443	0.035042	4.692404	5.26E-06	3492

e43	E.1.1	E.2.1	E.3.4	E.4.2	E.5.1	E.6.1	0.16443	0.035042	4.692404	5.26E-06	3492
e52	E.1.1	E.2.2	E.3.1	E.4.1	E.5.2	E.6.1	0.164699	0.021805	7.553413	1.92E-12	3492
e298	E.1.1	E.2.2	E.3.2	E.4.1	E.5.1	E.6.4	0.164845	0.037733	4.368747	2.08E-05	3492
e64	E.1.1	E.2.2	E.3.3	E.4.1	E.5.2	E.6.1	0.165095	0.022027	7.495168	2.7E-12	3492
e295	E.1.1	E.2.1	E.3.2	E.4.1	E.5.1	E.6.4	0.166722	0.042627	3.911203	0.000129	3492
e268	E.1.1	E.2.2	E.3.1	E.4.2	E.5.2	E.6.3	0.169347	0.027891	6.071661	7.52E-09	3291
e280	E.1.1	E.2.2	E.3.3	E.4.2	E.5.2	E.6.3	0.169676	0.027587	6.150608	5E-09	3291
e301	E.1.1	E.2.1	E.3.3	E.4.1	E.5.1	E.6.4	0.169839	0.034949	4.859686	2.51E-06	3492
e7	E.1.1	E.2.1	E.3.2	E.4.1	E.5.1	E.6.1	0.172349	0.043727	3.941474	0.000115	3492
e13	E.1.1	E.2.1	E.3.3	E.4.1	E.5.1	E.6.1	0.174083	0.034178	5.09339	8.66E-07	3492
e1	E.1.1	E.2.1	E.3.1	E.4.1	E.5.1	E.6.1	0.17506	0.03054	5.73223	3.99E-08	3492
e253	E.1.1	E.2.1	E.3.3	E.4.1	E.5.2	E.6.3	0.175539	0.027116	6.473689	8.44E-10	3490
e28	E.1.1	E.2.2	E.3.1	E.4.2	E.5.1	E.6.1	0.177985	0.032938	5.403605	2.09E-07	3291
e241	E.1.1	E.2.1	E.3.1	E.4.1	E.5.2	E.6.3	0.178991	0.026931	6.646302	3.3E-10	3490
e22	E.1.1	E.2.2	E.3.4	E.4.1	E.5.1	E.6.1	0.179016	0.03893	4.598443	7.9E-06	3492
e46	E.1.1	E.2.2	E.3.4	E.4.2	E.5.1	E.6.1	0.179016	0.03893	4.598443	7.9E-06	3492
e40	E.1.1	E.2.2	E.3.3	E.4.2	E.5.1	E.6.1	0.181016	0.032889	5.503862	1.29E-07	3291
e283	E.1.1	E.2.1	E.3.4	E.4.2	E.5.2	E.6.3	0.181719	0.033147	5.482272	1.43E-07	3356
e292	E.1.1	E.2.2	E.3.1	E.4.1	E.5.1	E.6.4	0.182613	0.031894	5.725654	4.13E-08	3492
e10	E.1.1	E.2.2	E.3.2	E.4.1	E.5.1	E.6.1	0.184605	0.040683	4.537652	1.02E-05	3492
e304	E.1.1	E.2.2	E.3.3	E.4.1	E.5.1	E.6.4	0.1885	0.032629	5.777068	3.19E-08	3492
e247	E.1.1	E.2.1	E.3.2	E.4.1	E.5.2	E.6.3	0.195496	0.032561	6.004029	1E-08	3490
e217	E.1.1	E.2.1	E.3.1	E.4.2	E.5.1	E.6.3	0.198876	0.043239	4.599461	8.04E-06	3291
e259	E.1.1	E.2.1	E.3.4	E.4.1	E.5.2	E.6.3	0.204248	0.031541	6.475644	8.35E-10	3490
e229	E.1.1	E.2.1	E.3.3	E.4.2	E.5.1	E.6.3	0.204832	0.044222	4.631868	7E-06	3291
e199	E.1.1	E.2.1	E.3.2	E.4.1	E.5.1	E.6.3	0.208886	0.049821	4.192724	4.28E-05	3490
e4	E.1.1	E.2.2	E.3.1	E.4.1	E.5.1	E.6.1	0.21034	0.031326	6.714571	2.27E-10	3492
e16	E.1.1	E.2.2	E.3.3	E.4.1	E.5.1	E.6.1	0.211051	0.032069	6.581143	4.72E-10	3492
e256	E.1.1	E.2.2	E.3.3	E.4.1	E.5.2	E.6.3	0.21756	0.028898	7.528553	2.22E-12	3490

e244	E.1.1	E.2.2	E.3.1	E.4.1	E.5.2	E.6.3	0.218163	0.028701	7.601243	1.45E-12	3490
e211	E.1.1	E.2.1	E.3.4	E.4.1	E.5.1	E.6.3	0.220498	0.040459	5.449882	1.6E-07	3490
e235	E.1.1	E.2.1	E.3.4	E.4.2	E.5.1	E.6.3	0.220498	0.040459	5.449882	1.6E-07	3490
e205	E.1.1	E.2.1	E.3.3	E.4.1	E.5.1	E.6.3	0.224766	0.045287	4.963164	1.57E-06	3490
e193	E.1.1	E.2.1	E.3.1	E.4.1	E.5.1	E.6.3	0.228535	0.040363	5.661932	5.67E-08	3490
e220	E.1.1	E.2.2	E.3.1	E.4.2	E.5.1	E.6.3	0.237964	0.043638	5.453182	1.64E-07	3291
e232	E.1.1	E.2.2	E.3.3	E.4.2	E.5.1	E.6.3	0.241799	0.043635	5.541446	1.07E-07	3291
e286	E.1.1	E.2.2	E.3.4	E.4.2	E.5.2	E.6.3	0.246099	0.034127	7.21131	1.55E-11	3356
e250	E.1.1	E.2.2	E.3.2	E.4.1	E.5.2	E.6.3	0.267808	0.032981	8.120156	6.53E-14	3490
e262	E.1.1	E.2.2	E.3.4	E.4.1	E.5.2	E.6.3	0.272091	0.033209	8.193311	4.19E-14	3490
e208	E.1.1	E.2.2	E.3.3	E.4.1	E.5.1	E.6.3	0.277969	0.042332	6.566482	5.11E-10	3490
e196	E.1.1	E.2.2	E.3.1	E.4.1	E.5.1	E.6.3	0.278378	0.041331	6.735324	2.03E-10	3490
e202	E.1.1	E.2.2	E.3.2	E.4.1	E.5.1	E.6.3	0.28007	0.042703	6.55858	5.33E-10	3490
e214	E.1.1	E.2.2	E.3.4	E.4.1	E.5.1	E.6.3	0.284599	0.041768	6.813787	1.31E-10	3490
e238	E.1.1	E.2.2	E.3.4	E.4.2	E.5.1	E.6.3	0.284599	0.041768	6.813787	1.31E-10	3490

Notes: Each row presents a re-analysis featured in Figure 1 testing the sensitivity of results for English test scores presented in Table 3 of Romero et al. (2020). The category for re-analysis type is discussed in Sub-section 3.2. The point estimates are in standard deviations.

Table A.6: Re-analyses Performed in the Specification Curve for Figure 2

(1) Combination Label	(2) Dimension Sample	(3) Dimension Treatment	(4) Dimension Controls	(5) Dimension Missing Obs.	(6) Dimension Model	(7) Dimension Outcome	(8) Treatment Effect	(9) SE	(10) t-value	(11) p-value	(12) Obs.
m469	M.1.1	M.2.1	M.3.3	M.4.2	M.5.2	M.6.5	0.028004	0.004121	6.795264	1.59E-10	3291
m475	M.1.1	M.2.1	M.3.4	M.4.2	M.5.2	M.6.5	0.028034	0.004345	6.452547	1.02E-09	3356
m457	M.1.1	M.2.1	M.3.1	M.4.2	M.5.2	M.6.5	0.028175	0.004341	6.489712	8.35E-10	3291
m439	M.1.1	M.2.1	M.3.2	M.4.1	M.5.2	M.6.5	0.031943	0.004245	7.525637	2.26E-12	3492
m445	M.1.1	M.2.1	M.3.3	M.4.1	M.5.2	M.6.5	0.03246	0.004358	7.448904	3.54E-12	3492
m118	M.1.1	M.2.2	M.3.4	M.4.1	M.5.1	M.6.2	0.033078	0.042621	0.7761	0.438726	3356
m142	M.1.1	M.2.2	M.3.4	M.4.2	M.5.1	M.6.2	0.033078	0.042621	0.7761	0.438726	3356
m451	M.1.1	M.2.1	M.3.4	M.4.1	M.5.2	M.6.5	0.033147	0.004613	7.186263	1.61E-11	3492
m433	M.1.1	M.2.1	M.3.1	M.4.1	M.5.2	M.6.5	0.034029	0.004731	7.192764	1.55E-11	3492
m460	M.1.1	M.2.2	M.3.1	M.4.2	M.5.2	M.6.5	0.036041	0.004534	7.94967	2.12E-13	3291
m472	M.1.1	M.2.2	M.3.3	M.4.2	M.5.2	M.6.5	0.036061	0.004456	8.093423	8.98E-14	3291
m478	M.1.1	M.2.2	M.3.4	M.4.2	M.5.2	M.6.5	0.03685	0.004643	7.936276	2.29E-13	3356
m409	M.1.1	M.2.1	M.3.1	M.4.2	M.5.1	M.6.5	0.036924	0.007301	5.05707	1.06E-06	3291
m421	M.1.1	M.2.1	M.3.3	M.4.2	M.5.1	M.6.5	0.037835	0.007197	5.257366	4.18E-07	3291
m403	M.1.1	M.2.1	M.3.4	M.4.1	M.5.1	M.6.5	0.040216	0.007921	5.077045	9.34E-07	3492
m427	M.1.1	M.2.1	M.3.4	M.4.2	M.5.1	M.6.5	0.040216	0.007921	5.077045	9.34E-07	3492
m442	M.1.1	M.2.2	M.3.2	M.4.1	M.5.2	M.6.5	0.040421	0.005126	7.885523	2.68E-13	3492
m454	M.1.1	M.2.2	M.3.4	M.4.1	M.5.2	M.6.5	0.040682	0.005205	7.816217	4.06E-13	3492
m448	M.1.1	M.2.2	M.3.3	M.4.1	M.5.2	M.6.5	0.040817	0.005144	7.934391	2E-13	3492
m436	M.1.1	M.2.2	M.3.1	M.4.1	M.5.2	M.6.5	0.041266	0.005211	7.919558	2.19E-13	3492
m391	M.1.1	M.2.1	M.3.2	M.4.1	M.5.1	M.6.5	0.041417	0.008363	4.952224	1.65E-06	3492
m397	M.1.1	M.2.1	M.3.3	M.4.1	M.5.1	M.6.5	0.041616	0.007096	5.864336	2.05E-08	3492
m385	M.1.1	M.2.1	M.3.1	M.4.1	M.5.1	M.6.5	0.04235	0.007176	5.901361	1.7E-08	3492
m115	M.1.1	M.2.1	M.3.4	M.4.1	M.5.1	M.6.2	0.043926	0.038506	1.140748	0.255516	3356

m139	M.1.1	M.2.1	M.3.4	M.4.2	M.5.1	M.6.2	0.043926	0.038506	1.140748	0.255516	3356
m406	M.1.1	M.2.2	M.3.4	M.4.1	M.5.1	M.6.5	0.047027	0.008658	5.431845	1.75E-07	3492
m430	M.1.1	M.2.2	M.3.4	M.4.2	M.5.1	M.6.5	0.047027	0.008658	5.431845	1.75E-07	3492
m106	M.1.1	M.2.2	M.3.2	M.4.1	M.5.1	M.6.2	0.04763	0.044454	1.071436	0.285432	3356
m394	M.1.1	M.2.2	M.3.2	M.4.1	M.5.1	M.6.5	0.048143	0.00876	5.496061	1.28E-07	3492
m412	M.1.1	M.2.2	M.3.1	M.4.2	M.5.1	M.6.5	0.048243	0.00715	6.747263	2.07E-10	3291
m424	M.1.1	M.2.2	M.3.3	M.4.2	M.5.1	M.6.5	0.048865	0.007153	6.831042	1.3E-10	3291
m388	M.1.1	M.2.2	M.3.1	M.4.1	M.5.1	M.6.5	0.05299	0.007415	7.146285	2.02E-11	3492
m400	M.1.1	M.2.2	M.3.3	M.4.1	M.5.1	M.6.5	0.052996	0.007385	7.176491	1.7E-11	3492
m121	M.1.1	M.2.1	M.3.1	M.4.2	M.5.1	M.6.2	0.054055	0.032658	1.655176	0.099661	3291
m97	M.1.1	M.2.1	M.3.1	M.4.1	M.5.1	M.6.2	0.055973	0.032441	1.725351	0.08621	3356
m133	M.1.1	M.2.1	M.3.3	M.4.2	M.5.1	M.6.2	0.060765	0.032607	1.863571	0.064038	3291
m103	M.1.1	M.2.1	M.3.2	M.4.1	M.5.1	M.6.2	0.06589	0.042571	1.54777	0.123464	3356
m109	M.1.1	M.2.1	M.3.3	M.4.1	M.5.1	M.6.2	0.066875	0.033417	2.00123	0.046895	3356
m100	M.1.1	M.2.2	M.3.1	M.4.1	M.5.1	M.6.2	0.07536	0.031128	2.420998	0.016488	3356
m124	M.1.1	M.2.2	M.3.1	M.4.2	M.5.1	M.6.2	0.075627	0.031371	2.410743	0.016943	3291
m136	M.1.1	M.2.2	M.3.3	M.4.2	M.5.1	M.6.2	0.079372	0.03205	2.476531	0.014206	3291
m112	M.1.1	M.2.2	M.3.3	M.4.1	M.5.1	M.6.2	0.081917	0.032363	2.531169	0.012239	3356
m73	M.1.1	M.2.1	M.3.1	M.4.2	M.5.2	M.6.1	0.111602	0.020805	5.364212	2.52E-07	3291
m91	M.1.1	M.2.1	M.3.4	M.4.2	M.5.2	M.6.1	0.112035	0.020445	5.479717	1.45E-07	3356
m85	M.1.1	M.2.1	M.3.3	M.4.2	M.5.2	M.6.1	0.113803	0.019886	5.722749	4.4E-08	3291
m67	M.1.1	M.2.1	M.3.4	M.4.1	M.5.2	M.6.1	0.139645	0.021862	6.387563	1.34E-09	3492
m55	M.1.1	M.2.1	M.3.2	M.4.1	M.5.2	M.6.1	0.142213	0.021122	6.733047	2.05E-10	3492
m265	M.1.1	M.2.1	M.3.1	M.4.2	M.5.2	M.6.3	0.142869	0.025876	5.521247	1.18E-07	3291
m49	M.1.1	M.2.1	M.3.1	M.4.1	M.5.2	M.6.1	0.143098	0.022148	6.460922	9.04E-10	3492
m61	M.1.1	M.2.1	M.3.3	M.4.1	M.5.2	M.6.1	0.143771	0.02103	6.836568	1.16E-10	3492
m277	M.1.1	M.2.1	M.3.3	M.4.2	M.5.2	M.6.3	0.14382	0.024756	5.809389	2.85E-08	3291
m76	M.1.1	M.2.2	M.3.1	M.4.2	M.5.2	M.6.1	0.146354	0.023393	6.25631	2.88E-09	3291
m88	M.1.1	M.2.2	M.3.3	M.4.2	M.5.2	M.6.1	0.148404	0.022895	6.48197	8.7E-10	3291

m94	M.1.1	M.2.2	M.3.4	M.4.2	M.5.2	M.6.1	0.148645	0.022996	6.464102	9.58E-10	3356
m25	M.1.1	M.2.1	M.3.1	M.4.2	M.5.1	M.6.1	0.153676	0.03471	4.427483	1.66E-05	3291
m37	M.1.1	M.2.1	M.3.3	M.4.2	M.5.1	M.6.1	0.161095	0.034366	4.687591	5.5E-06	3291
m379	M.1.1	M.2.1	M.3.4	M.4.2	M.5.2	M.6.4	0.163571	0.025648	6.377647	1.52E-09	2688
m361	M.1.1	M.2.1	M.3.1	M.4.2	M.5.2	M.6.4	0.164595	0.025522	6.449041	1.04E-09	2646
m70	M.1.1	M.2.2	M.3.4	M.4.1	M.5.2	M.6.1	0.166824	0.025546	6.530244	6.22E-10	3492
m373	M.1.1	M.2.1	M.3.3	M.4.2	M.5.2	M.6.4	0.16813	0.024698	6.807527	1.48E-10	2646
m58	M.1.1	M.2.2	M.3.2	M.4.1	M.5.2	M.6.1	0.170244	0.025493	6.677944	2.78E-10	3492
m19	M.1.1	M.2.1	M.3.4	M.4.1	M.5.1	M.6.1	0.172078	0.037943	4.535199	1.04E-05	3492
m43	M.1.1	M.2.1	M.3.4	M.4.2	M.5.1	M.6.1	0.172078	0.037943	4.535199	1.04E-05	3492
m283	M.1.1	M.2.1	M.3.4	M.4.2	M.5.2	M.6.3	0.172833	0.031472	5.491691	1.37E-07	3356
m52	M.1.1	M.2.2	M.3.1	M.4.1	M.5.2	M.6.1	0.17304	0.025922	6.675458	2.82E-10	3492
m64	M.1.1	M.2.2	M.3.3	M.4.1	M.5.2	M.6.1	0.175395	0.025792	6.800324	1.41E-10	3492
m253	M.1.1	M.2.1	M.3.3	M.4.1	M.5.2	M.6.3	0.175848	0.025756	6.827367	1.22E-10	3490
m343	M.1.1	M.2.1	M.3.2	M.4.1	M.5.2	M.6.4	0.176093	0.025336	6.950367	6.12E-11	2781
m241	M.1.1	M.2.1	M.3.1	M.4.1	M.5.2	M.6.3	0.178371	0.02732	6.529016	6.26E-10	3490
m1	M.1.1	M.2.1	M.3.1	M.4.1	M.5.1	M.6.1	0.182026	0.033525	5.429631	1.77E-07	3492
m349	M.1.1	M.2.1	M.3.3	M.4.1	M.5.2	M.6.4	0.183029	0.026223	6.979817	5.19E-11	2781
m355	M.1.1	M.2.1	M.3.4	M.4.1	M.5.2	M.6.4	0.183538	0.027391	6.700767	2.45E-10	2781
m7	M.1.1	M.2.1	M.3.2	M.4.1	M.5.1	M.6.1	0.185011	0.040805	4.534068	1.04E-05	3492
m295	M.1.1	M.2.1	M.3.2	M.4.1	M.5.1	M.6.4	0.185337	0.045856	4.041694	7.79E-05	2781
m13	M.1.1	M.2.1	M.3.3	M.4.1	M.5.1	M.6.1	0.186132	0.033686	5.525504	1.11E-07	3492
m307	M.1.1	M.2.1	M.3.4	M.4.1	M.5.1	M.6.4	0.187683	0.047192	3.977025	0.0001	2781
m331	M.1.1	M.2.1	M.3.4	M.4.2	M.5.1	M.6.4	0.187683	0.047192	3.977025	0.0001	2781
m247	M.1.1	M.2.1	M.3.2	M.4.1	M.5.2	M.6.3	0.190087	0.030066	6.322322	1.9E-09	3490
m268	M.1.1	M.2.2	M.3.1	M.4.2	M.5.2	M.6.3	0.191141	0.028724	6.654308	3.43E-10	3291
m280	M.1.1	M.2.2	M.3.3	M.4.2	M.5.2	M.6.3	0.192351	0.028203	6.820138	1.39E-10	3291
m337	M.1.1	M.2.1	M.3.1	M.4.1	M.5.2	M.6.4	0.19287	0.028322	6.809808	1.34E-10	2781
m217	M.1.1	M.2.1	M.3.1	M.4.2	M.5.1	M.6.3	0.19531	0.043163	4.524897	1.11E-05	3291

m259	M.1.1	M.2.1	M.3.4	M.4.1	M.5.2	M.6.3	0.197249	0.031469	6.267959	2.53E-09	3490
m382	M.1.1	M.2.2	M.3.4	M.4.2	M.5.2	M.6.4	0.197752	0.030667	6.448346	1.04E-09	2688
m22	M.1.1	M.2.2	M.3.4	M.4.1	M.5.1	M.6.1	0.198193	0.042228	4.693376	5.23E-06	3492
m46	M.1.1	M.2.2	M.3.4	M.4.2	M.5.1	M.6.1	0.198193	0.042228	4.693376	5.23E-06	3492
m364	M.1.1	M.2.2	M.3.1	M.4.2	M.5.2	M.6.4	0.199106	0.028508	6.984243	5.57E-11	2646
m313	M.1.1	M.2.1	M.3.1	M.4.2	M.5.1	M.6.4	0.202305	0.041656	4.856544	2.62E-06	2646
m229	M.1.1	M.2.1	M.3.3	M.4.2	M.5.1	M.6.3	0.202748	0.042538	4.766294	3.9E-06	3291
m376	M.1.1	M.2.2	M.3.3	M.4.2	M.5.2	M.6.4	0.204157	0.028551	7.150652	2.19E-11	2646
m28	M.1.1	M.2.2	M.3.1	M.4.2	M.5.1	M.6.1	0.205061	0.034993	5.860044	2.21E-08	3291
m310	M.1.1	M.2.2	M.3.4	M.4.1	M.5.1	M.6.4	0.205962	0.048729	4.226649	3.73E-05	2781
m334	M.1.1	M.2.2	M.3.4	M.4.2	M.5.1	M.6.4	0.205962	0.048729	4.226649	3.73E-05	2781
m10	M.1.1	M.2.2	M.3.2	M.4.1	M.5.1	M.6.1	0.207014	0.042887	4.826968	2.9E-06	3492
m298	M.1.1	M.2.2	M.3.2	M.4.1	M.5.1	M.6.4	0.20793	0.047948	4.336558	2.38E-05	2781
m325	M.1.1	M.2.1	M.3.3	M.4.2	M.5.1	M.6.4	0.208747	0.041267	5.058414	1.05E-06	2646
m40	M.1.1	M.2.2	M.3.3	M.4.2	M.5.1	M.6.1	0.210019	0.034929	6.012721	1.02E-08	3291
m199	M.1.1	M.2.1	M.3.2	M.4.1	M.5.1	M.6.3	0.214969	0.043077	4.990385	1.39E-06	3490
m346	M.1.1	M.2.2	M.3.2	M.4.1	M.5.2	M.6.4	0.218078	0.030838	7.071693	3.09E-11	2781
m211	M.1.1	M.2.1	M.3.4	M.4.1	M.5.1	M.6.3	0.21813	0.042228	5.165588	6.2E-07	3490
m235	M.1.1	M.2.1	M.3.4	M.4.2	M.5.1	M.6.3	0.21813	0.042228	5.165588	6.2E-07	3490
m358	M.1.1	M.2.2	M.3.4	M.4.1	M.5.2	M.6.4	0.2192	0.031391	6.982814	5.1E-11	2781
m244	M.1.1	M.2.2	M.3.1	M.4.1	M.5.2	M.6.3	0.220947	0.031723	6.964932	5.64E-11	3490
m256	M.1.1	M.2.2	M.3.3	M.4.1	M.5.2	M.6.3	0.221985	0.031639	7.016294	4.22E-11	3490
m193	M.1.1	M.2.1	M.3.1	M.4.1	M.5.1	M.6.3	0.22684	0.041644	5.447065	1.63E-07	3490
m205	M.1.1	M.2.1	M.3.3	M.4.1	M.5.1	M.6.3	0.228067	0.041307	5.521238	1.13E-07	3490
m301	M.1.1	M.2.1	M.3.3	M.4.1	M.5.1	M.6.4	0.228705	0.041081	5.567222	9.05E-08	2781
m352	M.1.1	M.2.2	M.3.3	M.4.1	M.5.2	M.6.4	0.229491	0.029894	7.676919	9.28E-13	2781
m4	M.1.1	M.2.2	M.3.1	M.4.1	M.5.1	M.6.1	0.22967	0.035522	6.465553	8.82E-10	3492
m340	M.1.1	M.2.2	M.3.1	M.4.1	M.5.2	M.6.4	0.231593	0.030331	7.635393	1.19E-12	2781
m16	M.1.1	M.2.2	M.3.3	M.4.1	M.5.1	M.6.1	0.233452	0.03552	6.572339	4.95E-10	3492

m289	M.1.1	M.2.1	M.3.1	M.4.1	M.5.1	M.6.4	0.236801	0.042962	5.511818	1.19E-07	2781
m286	M.1.1	M.2.2	M.3.4	M.4.2	M.5.2	M.6.3	0.25277	0.035413	7.137722	2.35E-11	3356
m316	M.1.1	M.2.2	M.3.1	M.4.2	M.5.1	M.6.4	0.254765	0.042015	6.063692	7.84E-09	2646
m328	M.1.1	M.2.2	M.3.3	M.4.2	M.5.1	M.6.4	0.260965	0.042253	6.176324	4.37E-09	2646
m220	M.1.1	M.2.2	M.3.1	M.4.2	M.5.1	M.6.3	0.264298	0.043308	6.102716	6.41E-09	3291
m250	M.1.1	M.2.2	M.3.2	M.4.1	M.5.2	M.6.3	0.267946	0.036525	7.336042	6.8E-12	3490
m232	M.1.1	M.2.2	M.3.3	M.4.2	M.5.1	M.6.3	0.269109	0.043091	6.245154	3.05E-09	3291
m262	M.1.1	M.2.2	M.3.4	M.4.1	M.5.2	M.6.3	0.27035	0.037146	7.277982	9.51E-12	3490
m304	M.1.1	M.2.2	M.3.3	M.4.1	M.5.1	M.6.4	0.28756	0.042353	6.789539	1.5E-10	2781
m292	M.1.1	M.2.2	M.3.1	M.4.1	M.5.1	M.6.4	0.288731	0.043091	6.700566	2.45E-10	2781
m214	M.1.1	M.2.2	M.3.4	M.4.1	M.5.1	M.6.3	0.291285	0.044954	6.479569	8.18E-10	3490
m238	M.1.1	M.2.2	M.3.4	M.4.2	M.5.1	M.6.3	0.291285	0.044954	6.479569	8.18E-10	3490
m196	M.1.1	M.2.2	M.3.1	M.4.1	M.5.1	M.6.3	0.291659	0.043973	6.632727	3.56E-10	3490
m202	M.1.1	M.2.2	M.3.2	M.4.1	M.5.1	M.6.3	0.291866	0.044655	6.535975	6.03E-10	3490
m208	M.1.1	M.2.2	M.3.3	M.4.1	M.5.1	M.6.3	0.29427	0.043795	6.71925	2.21E-10	3490

Notes: Each row presents a re-analysis featured in Figure 2 testing the sensitivity of results for mathematics test scores presented in Table 3 of Romero et al. (2020). The category for re-analysis type is discussed in Sub-section 3.2. The point estimates are in standard deviations.